Reading Between the Lines: How People Perceive and Detect Cultural Erasure in AI-Retold Bengali Folktales

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
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Month Year.

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Declaration

It is hereby declared that

1. The thesis submitted is my/our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Ethics Statement

All the members of the group have participated in the Collaborative Institutional Training Initiative (CITI) on the ethics and implications of human research and have acquired certifications upon completion.

This study will be conducted with consenting adult participants and public domain resources. Participants will be informed about their rights to withdraw at any time during the study.

No identifiable information about the participants will be stored. All data gathered from them will be encrypted and stored with restricted access to one research member of the group.

This study was reviewed and approved by the Institutional Review Board (IRB) of BRAC University to ensure compliance with ethical research standards.

Abstract

As LLMs emerge as foundational digital interfaces via which people around the globe access, share, and experience both historical and cultural narratives, there emerges an important sociotechnical challenge for Human-Computer Interaction (HCI) research: Are human users able to tell when the very specific and localized cultural heritage has been manipulated, removed, or substituted by machine-generated renditions? This thesis tackles the above question with the example of Bengali folktales, which are an essential repository of intangible cultural heritage of Bangladesh, and carefully analyzes the link between automated audit of cultural manipulations and humans' perceptual ability to notice such alterations.

We contribute to the intersection of computing, digital humanities, and human-centered AI in three related methodological, empirical, and pragmatic ways.

First, we establish a verified and expertly annotated Reference Corpus of 27 Bengali folktales (resulting in 54 story pairs of Bengali and English texts), each systematically mapped for culturally non-substitutable lexical items, structural atomicity of plot points, socio-cultural behavioral traits, and clear structural impositions of the Western narrative form.

Second, we develop a multi-component automated evaluation pipeline architecture. While going beyond the traditional string matching and vector embeddings paradigms, known empirically to be unable to identify deeper cultural shifts, our automated system implements a comprehensive multi-criteria rubric-based evaluation layer, enabling us to define a novel structural indicator of cultural degradation: Structural Displacement Indicator (SDI). Testing of 1,296 stories generated by six state-of-the-art LLMs (GPT-4o, Gemini 2.0 Flash, Llama-3-70B, Qwen-2.5-72B among others) across four different prompt engineering techniques demonstrates that localized Bengali folktales undergo systematic cultural compression, with automatic transformation into Westernized, Disneyfied, or Eurocentric forms of storytelling.

Finally, we design and conduct a human user verification experiment ($n = 76$) aimed at measuring human users' ability to detect above machine distortions. Our human study results in an interesting cognitive effect, indicating that in standard blind-reading conditions, human subjects are plagued by overconfidence bias, achieving only near-chance accuracy in telling high-quality machine distortions from authentic stories based almost exclusively on stylistic superficial fluency of stories. By contrast, when provided with dynamically generated evaluations and feedback from the automated analysis pipeline, humans achieve a calibrated detection accuracy of 86.0%. These findings suggest that while automated cultural audits alone fall short, their combination with humans is particularly powerful in terms of human cognitive scaffolding. Based on these insights, we derive six HCI design guidelines for building culturally accountable generative systems, demonstrating how structural transparency, fail-safe prompt architectures, and human-in-the-loop validation can mitigate digital neocolonialism and protect vulnerable, low-resource cultural heritage from automated erasure.

Keywords: *Human-Computer Interaction (HCI), Cultural Erasure, Large Language Models (LLMs), Bengali Folktales, Intangible Cultural Heritage, Structural Displacement Indicator (SDI), Deceptive Fluency, Human-AI Alignment, Digital Humanities.*

Dedication

To the grandmothers, community narrators, and oral historians of Bengal who have preserved the fragile yet colorful fabric of our intangible culture through generations; and to the computer scientists, designers, and researchers of the Global South in the coming era whose responsibility will be to safeguard the unique and irreplaceable aspects of our heritage in the digital era.

Acknowledgement

All praise is due to Allah, who blesses us with everything we have. He has given us strength and facilitated each step of this work, overcoming all challenges, academic or technical.

Firstly, we would like to thank our thesis advisor, **Dr. Taiabul Haque**, Professor, Department of Computer Science and Engineering, BRAC University, for his constant encouragement and support. His intellectual stimulation, stringent standards, and thoughtful advice urged us to expand this multidisciplinary work beyond a machine learning project centered around local dataset analysis to a holistic sociotechnical research worthy of publication. He inspired us and believed in us, giving us reasons to conduct this research for a greater cause of preserving local heritage with computers.

Our mentor **Md. Rezuwan Hassan**, Research Assistant to our advisor, deserves our many thanks. His diligent low-level technical support, constructive comments on our engineering workflow, and timely advice on data preprocessing, scripting, and evaluation metrics were crucial to the development of this thesis. He patiently answered our never-ending queries and did not hesitate to point out our mistakes. Thanks to him, we were able to validate our claims.

We would like to thank our parents from the bottom of our hearts. They have given us everything, including their unconditional love, patience, and sacrifices. We could not have asked for better parents.

Lastly, we want to thank all the participants who gave us their time and helped us with our user study. Specifically, the community members and expert annotators who helped us annotate the corpus, and the human participants who helped us evaluate our models. Thank you for allowing us to use your cultural expertise and focus. We hope this thesis benefits the preservation of Bengali folktales.

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Chapter 1

Introduction

1.1 Background

Bengali folktales constitute one of the oldest and most culturally dense bodies of intangible heritage in South Asia. Transmitted across generations through oral performance, manuscript tradition, and community storytelling, these narratives encode cosmological beliefs, moral frameworks, kinship obligations, and regional identities that cannot be reduced to their surface plot content. The specific named entities that populate Bengali folklore — the *Rakkhosh* ogre-demon, the *Yaksha* spirit, the *Bastu Devi* household deity, the river goddess of the Padma — carry cultural meaning that generic Western equivalents such as "demon," "spirit," "household deity," and "river spirit" cannot preserve. This irreducibility is not incidental to the folklore; it is constitutive of it. When these specific terms are replaced, the narrative does not merely lose a word — it loses the cosmological framework in which the word has meaning.

The digitization and computational mediation of cultural heritage has emerged as a critical site of both opportunity and risk. Xanthaki [28], in a technical report commissioned by the United Nations Office of the High Commissioner for Human Rights, identifies five criteria for culturally responsible heritage digitization: community consent, contextual integrity, cultural sovereignty, interpretive fidelity, and sustainability of access. The proliferation of large language models (LLMs) as interfaces through which communities encounter, share, and generate cultural content creates new pressures on each of these criteria, particularly interpretive fidelity. Contemporary LLMs are known to exhibit systematic biases toward culturally dominant — predominantly Western, English-speaking — narrative conventions, linguistic registers, and conceptual frameworks [20], [22]. These biases emerge from the structural imbalance of large-scale pre-training corpora, in which Western cultural content overwhelmingly dominates [5], and manifest across languages and tasks as a tendency to interpret culturally ambiguous inputs through the lens of Western default assumptions [30], [31].

Bengali presents a particularly acute case of this structural disadvantage. Despite being spoken by a substantial portion of the world's population across Bangladesh, West Bengal, and diaspora communities globally, Bengali remains substantially underrepresented in the training data of widely deployed language models relative to

English, Mandarin, and other high-resource languages [5], [21]. This underrepresentation has measurable consequences: Billah et al. [9] demonstrate that NLP systems operating on Bengali folkloric text achieve classification accuracies of 88.56% on structural features, while noting the near-complete absence of culturally annotated Bengali datasets suitable for semantic evaluation tasks. Ayman et al. [24] further document the inadequacy of existing Bengali NLP resources in capturing dialectal and register diversity essential to faithful representation of regional oral traditions.

Against this backdrop, the deployment of LLMs in cultural content generation, literary assistance, and educational material production creates conditions of compounded risk for Bengali folkloric heritage. When AI systems trained on imbalanced data retell Bengali folktales — whether for educational platforms, digital storytelling applications, or conversational AI interfaces — they do not generate neutral retellings: they generate retellings shaped by the Western cultural defaults encoded in their training. Agarwal et al. [23] provide the most direct empirical evidence for this risk in an adjacent domain: in a 118-participant study, they demonstrate that AI writing assistance tools measurably shift user output toward Western stylistic norms even when users consciously attempt to preserve culturally distinctive voices. Applied to AI retelling of folklore rather than AI writing assistance, this homogenization effect has more severe consequences: the source tradition has no ongoing human author to compensate for the model's cultural defaults, and the output may circulate as an authoritative retelling without any indication that it represents a culturally distorted version of the source material.

1.2 Rational of the Study or Motivation

There are three main research gaps motivating this project that have been documented in isolation, but not in conjunction with one another for the case of Bengali folklore.

First, it is important to note the limitations of the current evaluation methods for measuring cultural fidelity in AI-generated narrative content. Evaluation tools based on standard NLP criteria, like BLEU score, cosine semantic similarity, and perplexity, measure surface form and proximity in meaning, not cultural authenticity [16]. The SocialCC benchmark, developed by Wu et al. [32], shows that it is impossible to assess cultural competence in language agents using static knowledge tests, highlighting the need for culturally grounded dynamic evaluation. In response, Montalan et al. [27] created Kalahi, a culturally curated tool for assessing the fidelity of Filipinos in narratives generated by language agents. There is no comparable cultural assessment tool for Bengali folkloric content at the present time, and this lack prevents any quantifiable assessment of AI-generated Bengali folklore.

The second research gap relates to the human dimension of the problem. As noted by Agarwal et al. [23] and Kambhamettu et al. [19], content generation by an AI influences user perception and engagement, but neither work considers the extent to which people familiar with a particular culture can identify cultural distortion induced by AI in heritage narrative content. It is not just an idle question, ei-

ther: Leger and Gutchess [8] find that cultural background influences both memory specificity and confidence calibration during narrative retrieval, and Cho et al. [26] further show that the interaction of cultural background with age impacts brain regions active during retrieval. Can Bengali speakers reliably recognize cultural distortions induced by AI-generated retellings? What conditions facilitate or interfere with successful recognition? This set of questions remains unstudied empirically.

Third, the postcolonial dimension of cultural AI mediation raises concerns about the epistemic dominance of Western perspectives. According to Risam [4], digital humanities infrastructure perpetuates Western epistemology, effectively marginalizing non-Western knowledge systems even as they purport to document them. Evidence of such epistemic inequality is provided directly by Chhikara et al. [25]: evaluating the performance of LLMs in answering culturally specific questions, the authors found average accuracy rates of only 41.6%. These results indicate that the accuracy of South Asian cultural knowledge encoded in LLMs lags considerably behind that for other cultures [4], [25]. Far from being an engineering problem to be resolved with additional training data, epistemic dominance issues point to the existence and perpetuation of systemic inequalities that structure the mediation of cultural heritage in the digital world [4], [28]. Documenting, measuring, and devising a culturally sensitive framework for addressing these inequalities thus becomes a research priority.

In sum, this research is framed as a human-computer interaction problem because the ultimate locus of cultural erasure is human communities, not computational systems. Cultural distortion in AI retelling of Bengali folktales is not just a technical challenge of designing a better language agent; it is an instance of representing Bengali culture to the people as an authorized text that distorts its essence. Conditions of such distortion must be analyzed, situations under which it becomes humanly imperceptible must be identified, and implications for designing systems that mediate cultural content must be explored.

1.3 Problem Statement

The alignment of three established trends — systematic cultural bias of contemporary LLMs [20], [22]; the structural underrepresentation of Bengali within AI training material [5], [9]; and the lack of culturally sensitive measurement frameworks for evaluating generated folklore [16], [27] — gives rise to a complex, interdisciplinary, and culturally urgent research problem, which this study aims to address by proposing three connected research questions:

RQ1: What types of cultural erasure are observed in contemporary LLMs' reiterations of Bengali folktales, and how are they affected by LLM model architecture, amount of information provided by a human user to an LLM (ranging from minimal name-only prompting to more involved source-text prompting), and language of human-AI interaction?

RQ2: Are native or heritage Bengali-speaking participants able to detect AI-generated retellings from expert-authored retellings of Bengali folktales? Which variables

— including the severity of automation-induced cultural erasure in a retelling, users' familiarity with Bengali folklore, their age, and their fluency in Bengali — mediate human participants' ability to detect such erasure?

RQ3: How do human-perceptual and Cultural Fidelity Perception Framework detection scores correlate? Is it feasible to apply the latter framework as a metric proxy of human cultural expertise?

In sum, the current research constitutes a diagnostics project, whose purpose is not to define cultural standards but rather to describe and quantify cultural erasure in AI-generated Bengali folklore and human detection thereof.

1.4 Objective

This research aims to achieve the following six goals:

1. **Create** a validated Reference Corpus consisting of 27 Bengali folktales selected from two main source corpora and annotated systematically on four fidelity scales with bilingual human-translated stories, which is used as the criterion against which AI-generated retellings can be evaluated.

2. **Develop** a systematic corpus of 1,296 AI-generated Bengali folktales across six large language models, four informational provision levels, 27 source texts, and two languages for user prompts.

3. **Identify** and describe different systematic phenomena related to the process of cultural distortion in the AI-generated stories, namely lexical replacement, structural displacement, and socio-cultural marker loss, as different types of erasure of culture.

4. **Design** and implement three independent approaches to evaluate culture in Bengali AI-generated folktales as a measure of cultural fidelity based on the principle of the highest agreement with human assessment, and select one of them for the main method for the experiment based on evidence-based design criteria.

5. **Perform** a Human Verification Study using 76 Bengali participants across two controlled scenarios of high cultural fidelity and low cultural fidelity AI-generated stories, and explore when people are able to detect the lack of cultural information in the stories.

6. **Evaluate** the degree of congruence between automated Cultural Fidelity Perception Framework scores and human perceptual detection of cultural erasure, and validate the framework itself as a potential proxy measure.

1.5 Methodology in Brief

This research involves three sequential stages, which will produce validated outputs that are then used as inputs to the following stage.

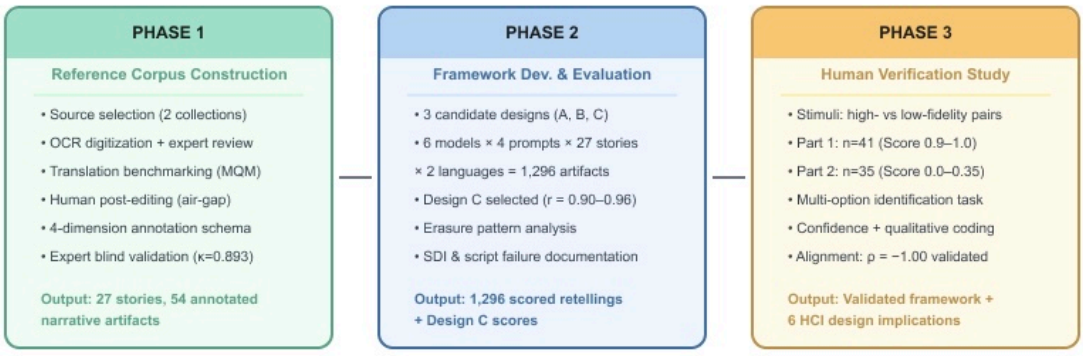


Figure 1.1: Three Phase Research Methodology Overview

Phase 1 – Reference Corpus Construction sets the standards on which evaluation would be done. First, 27 Bengali folktales are digitized using an Optical Character Recognition (OCR) system that is checked by an expert. They are translated from Bengali to English using a pipeline whose quality is benchmarked using various translation systems and then annotated in four cultural fidelity dimensions by folklore experts.

Phase 2 – Framework Development and Automated Evaluation will involve developing three independent candidate design ideas for measuring cultural fidelity. Afterward, we use the chosen design idea to evaluate the cultural fidelity of all 1,296 AI-generated versions and get a dataset in which the erasure pattern is characterized. Finally, we will compare three design ideas empirically to reference human judgments and choose the design showing better alignment with human judgments as the primary evaluator.

Phase 3 – Human Verification Study consists of two independent study conditions where 76 speakers of the Bengali language evaluate their perceptions regarding the cultural content produced by the AI.

1.6 Scopes and Challenges

Scope. There are four distinct boundaries that define the conditions within which the findings of this research can be generalized. The first boundary involves a restriction on the cultural scope to Bengali-language folklore. Specifically, findings relate to narrative traditions found in only two specific primary sources: the *Bangladesher Lokoj Shongskiti Gronthomala*, which is a series of multi-volume anthologies of oral folk tradition published by the Bangla Academy, and *Thakurmar Jhuli*, which is a canonical literary anthology of Bengali folk narratives. Patterns of cultural erasure described in the narrative traditions found in these two sources cannot be applied to other regional oral folklore traditions in Bangladesh, nor to West Bengali variants, or even other South Asian folklore traditions, without further confirmation. The second boundary relates to the way in which language models were studied. No training, fine-tuning, or other parameters were changed in any way. This was done to reflect the practical fact that most users interact with large language models using API endpoints without modifications to any parameters. It also means that results

presented in this research represent behavior exhibited by these models as is, rather than as potentially adjusted. The third boundary deals with cultural authenticity in this study. Specifically, cultural authenticity in this case was defined as adherence to one reference interpretation for each of the stories as validated by cultural experts. The framework recognizes that Bengali folklore is a diverse entity — there are different regional, performative, or generational variants of a particular narrative that are all equally valid. Finally, the Human Verification Study was limited by the use of online surveying tools, meaning that the study sample was made up of people who had access to internet and were capable and willing to participate in an online text-based experiment.

Challenges. One of the intellectual challenges in this research involved defining a method for measuring cultural authenticity. Specificity of folklore traditions is inherently multidimensional, subjective, and contestable to a degree that precludes the development of any simple metrics for measuring culture specificity. The solution of designing three independent measures for evaluation and selecting among them through validation against human judgment was selected based on the understanding that there can be no adequate technical solution without prior validation. The failure of Design B — embedding semantic similarity — to distinguish between culturally faithful and culturally distorted texts, while still generating a sufficiently high similarity score, demonstrated why this validation-first approach was necessary — without clear behavioral evidence that a measure tracks human cultural perception, a high metric score guarantees nothing about cultural quality.

Another challenge involved the construction of corpora. Bengali optical character recognition engines at the time of conducting the study failed to properly recognize traditional orthographic forms, conjunct consonants (*juktakkhar*), and different dialectal spelling conventions used in the original folklore texts. An absolute zero tolerance quality control standard was used in this study, where each recognized character was compared against the original document and manually verified by an expert, despite this increasing the amount of work compared to automated correction. This standard was necessary to make sure that Reference Corpus faithfully represents the folklore traditions instead of approximating them through poor OCR output. Access to a primary corpus was obtained after obtaining explicit permission directly from the Bangla Academy — this created the foundation for the cultural legitimacy of the study but is uncommon practice in typical NLP corpus construction processes.

A final challenge involved the recruitment process for the Human Verification Study. In order to create a sample of participants representing three age groups and three levels of familiarity with folklore, outreach needed to be done simultaneously via internal networks at BRAC University, community organizations, and various social media channels. The voluntary nature of this process, without financial incentives, required active engagement within the participant communities.

1.7 Team Overview

The study was conducted by a group of three members from the Department of Computer Science and Engineering, BRAC University, under the guidance of Dr. Taiabul Haque, Professor.

Sidratul Montaha (Student ID: 21201444) was the main researcher, having complete intellectual and practical control over the research. Her duties included research problem formulation, research theory formation, creation of the corpus at all the pipeline levels, development of the annotation scheme, Cultural Fidelity Perception Framework implementation on all the three models being considered, automatic evaluation process, Human Verification Study design and implementation, and all results analysis and report writing.

Nahita Nahian (Student ID: 24241175) and **Zead Raihan** (Student ID: 24241250) were the assisting researchers. They helped mainly in conducting literature reviews, entry of annotations into spreadsheets, and participant search for the Human Verification Study recruitment period. Also, as central figures in validating the human alignment component, they carefully reviewed, audited, and scored 40 completely developed, multi-tiered narrative generations so as to develop inter-rater alignment parameters and to evaluate the diagnostic sensitivity of the component rubrics used to evaluate the narrative generation streams.

1.8 Key Learnings and Insights

In summary, this research has produced results in the technical, academic, and methodological domains that go far beyond the specific lessons learned about Bengali folklore.

From the technical perspective, this research has shown that a lack of direct monetary expense does not necessarily mean a lack of methodological rigor — the entire pipeline, from language model inference and embeddings to OCR and cultural auditing, was performed using free-tier APIs and open-source software without any loss of the reliability of findings. The importance of this is clear for scholars who work in resource-limited settings, such as low-budget universities, and who have the biggest research interest in culturally non-dominant traditions, where computational power is at its minimum. This research has proven that embedding-based semantically similarity metrics, while being ubiquitous in natural language processing, not only fail to provide precision but may also produce misleading results when used in evaluating cultural similarity — a very high cosine similarity score may correlate with the total replacement of the culture, like the case of Design B shows.

From the academic point of view, the research has shown the importance of behavioral human validation for evaluating the culture. The correlation coefficient of Spearman $\rho = -1.00$ between framework results and behavioral human validation score was achieved as a result of empirical research and cannot be seen as guaranteed by the design. In terms of epistemology, the right sequence for cultural-sensitive artificial intelligence evaluation is to design, implement, and validate behaviorally.

Finally, from the methodological point of view, the requirement for receiving formal permission to perform the experiment from the Bangla Academy and involving folklore experts for annotation and validation of the results brought community involvement into the procedure, instead of just mentioning it rhetorically.

Chapter 2

Literature Review

2.1 Preliminaries

This literature review has been conducted on four online databases, namely ACM Digital Library, IEEE Xplore, Google Scholar, and Semantic Scholar, with the combination of keywords in relation to *cultural bias in language models*, *Bengali NLP*, *digitizing folklore*, *AI evaluation criteria*, *postcolonial digital humanities*, and *human cultural memory*. Around 50 papers have been initially selected by looking at the titles and abstracts; out of those, 30 articles have been included for full-text review since they are directly relevant to one or more of the research questions. The scope of the literature search includes papers published between 2016–2025, focusing especially on the past two years since LLMs' rapid implementation. The literature will be presented by eight themes mentioned below, each of which will be both methodologically informing and gap-identifying for this research study.

2.2 Review of Existing Research

Default Cultural Bias in Large Language Models. A convergence of recent empirical studies demonstrates that current LLMs show systematic tendencies toward cultural dominance in favor of dominant cultures, most notably of Western and English-language society. In the first line, Wang et al. [22] introduce SeaEval, a multi-lingual and multi-cultural evaluation suite which shows that top performing LLMs show significantly lower performance in Southeast Asian contexts compared to English-language tasks, and particularly when dealing with culturally specific reasoning and questions that require insider understanding. Kharchenko et al. [20] show that cultural bias is manifest in LLMs not only in errors and facts that require correction, but in the tendency to adopt culturally dominant frames when attempting to understand concepts with culture-specific meaning. Additionally, Salnikov et al. [30] show that cultural bias takes the form of geopolitical bias — that LLM outputs systematically represent Western-aligned countries in a more positive light in open-ended generation tasks. Finally, Sukiennik et al. [31] confirm that even when cultural frames are provided as prompts, LLMs fail to generate outputs aligned with non-Western cultural value sets. These works collectively demonstrate that default Western cultural behavior is a feature of LLM architecture, not an incidental byproduct, motivating further investigation into the effects of this problem in

the field of narrative retelling.

Disadvantages of Low-Resource Languages in NLP and Bengali Text Resources. The threat of cultural dominance in low-resource language contexts like Bengali is exacerbated by the disadvantaged position of such languages in current NLP development practices. Hovy and Prabhumoye [5] identify data imbalance in training data as one of five primary sources of bias in NLP systems, positing that models trained largely on high-resource language data embody the assumptions of that data as defaults. Indeed, Kakarla et al. [21] show a degradation in multilingual model performance when the language changes from a high-resource to a low-resource context in a variety of tasks, with degradation correlating with training data volume disparity. Specifically in the Bengali case, Billah et al. [9] conduct a systematic NLP analysis of Bengali folklore texts, producing classification results with 88.56% accuracy on genre and structural features and citing the lack of culturally authentic information in Bengali text collections as motivation to build such a reference corpus as part of this research project. Finally, Ayman et al. [24] develop BanglaBlend, a Bengali text corpus that distinguishes between two major variants of Bengali — *sadhubhasha* (high style) and *cholitbhasha* (low style) — confirming the lack of diversity in culturally-relevant corpora for Bengali-language content evaluation.

The Problem of Measurement in Cultural Evaluation Frameworks. A fundamental methodological issue that this body of research highlights is the inadequacy of existing metrics for evaluating cultural fidelity in AI-generated texts. Evaluation metrics based on semantic proximity such as BLEU, cosine similarity, and perplexity only account for surface or semantical similarities, leaving room for important differences that can indicate cultural biases or distortions of culturally authentic content. To address this issue, Montalan et al. [27] develop Kalahi, a cultural evaluation suite for Filipino texts demonstrating that LLM cultural knowledge cannot be measured by traditional benchmarks and must be evaluated using culturally designed test suites. SocialCC [32] offers a culturally oriented interactive evaluation benchmark for language agents demonstrating that static measures of knowledge do not capture cultural reasoning abilities of language models. Fragiadakis et al. [16] review existing evaluation methodologies in human-AI collaboration and note that existing frameworks prioritize completing the task over maintaining cultural authenticity — especially in the cultural content generation contexts. This convergence of recent works makes it clear that creating a comprehensive evaluation methodology for cultural fidelity is already a contribution, not just a necessary step.

Digital Folklore Corpus Construction and Preservation. Building folkloric corpora based on low-resource languages poses its own difficulties that the corpus construction literature attempts to address. Burda-Lassen [10] explores the issues associated with developing a Ukrainian-English parallel folktale corpus for low-resource machine translation, discussing issues like cold start pipeline design in the absence of any existing resources, an issue that directly relates to Bengali folklore translation efforts. Sun [13] explores cultural and communicative aspects of literary translation, explaining how machine translation flattens the unique cultural knowledge encoded in the process of translating stories from their original language.

Kwiecien et al. [7] develop a metadata schema for documenting Mekong River Basin folktales, which includes information about genres, regions, and structural elements of each tale, informing the metadata schema proposed in this thesis. Keersmaekers [6] explores methodological challenges in creating diverse multi-layered corpora for digitally preserving historical text traditions, an issue which also directly informs the design choices made for this research project.

Cultural Implications of HCI and Design Decisions. HCI implications of cultural homogenization in AI applications have begun to receive empirical examination. Specifically, Agarwal et al. [23] run a 118-participant experiment demonstrating that AI-based writing suggestion applications tend to make their users adopt Western-style stylistics, even when participants intentionally strive to maintain cultural uniqueness in their writing. The discovery that AI suggestions reduce cultural diversity in writing as an unintended consequence of tool usage, as opposed to a purposeful decision by users, is the most relevant precedent of this work in the domain of narrative retelling. Kambhamettu et al. [19] propose traceable text systems that reveal the origins of AI-generated content to the user as a solution to promoting informed AI usage. Huang et al. [17] explore explainability of AI-based interaction systems, showing that explainability methods, combined with ethnography research, produce valid assessments of AI behavior. Together, these works outline culturally-aware design as an active research topic and offer relevant methodological precedents for this thesis.

Digital Humanities and Postcolonial Studies. The general theoretical context for this research effort is represented by postcolonial studies of digital humanities. Risam [4] outlines epistemological assumptions built into digital infrastructure and argues that despite efforts to preserve non-Western cultures digitally, the very nature of those infrastructures leads to their systematic exclusion. Ray [3] and Mackey et al. [12] study legal and technological frameworks for indigenous data sovereignty, concluding that indigenous communities maintain ownership and definitional authority over digitally preserved material regardless of its intended use. As far as the problem statement of this project is concerned, relevant evidence is also provided by Chhikara et al. [25]: conducting an extensive analysis of subcultures of India, they conclude that LLMs exhibit 41.6% accuracy for answering culturally specific questions about them, thus demonstrating poor understanding of South Asian culture. Xanthaki [28], in her report for UN Office of the High Commissioner for Human Rights, proposes five criteria for culturally sensitive digitization of heritage material: consent, contextual integrity, and cultural sovereignty among them. Finally, Islam and Rahman [18] and Oveh et al. [29] study South Asian and African digital efforts at cultural preservation, respectively, demonstrating the need for culturally sensitive AI-based heritage preservation techniques.

Cross-Cultural Differences in Human Memory and Narrative Perception. An empirical literature that informs the human study of this research effort concerns cross-cultural and age-related differences in cognitive phenomena, particularly in memory encoding and perception of narratives. Leger and Gutchess [8] demonstrate that cultural background has a significant impact on the encoding of a story and subsequent memory of it, predicting cultural differences in performance when

detecting distortions to AI-based narrative retellings. Cho et al. [26] investigate age-related and cultural modulation of neural substrates involved in the process of memory retrieval, finding that age and cultural background affect both accuracy and confidence in narrative recall tasks. Taken together, these works outline theoretical reasons for age stratification in the human study of this research and the possibility that cultural perception performance will be affected by the interaction of age with folklore familiarity and distortions.

Propp's Structural Analysis of Folk Narratives and Computational Folklore. A literature that immediately relates to the method of Proppian Inversion developed in this project comes from computational folklore and literary studies. Originally, Propp (1928/1968) created and popularized the method of structural analysis of folk narratives, discovering 31 functions in Russian folk tales and establishing the methodology as a systematic research technique. In the decades following his original publication, the idea was tested in cross-cultural contexts, demonstrating systematic variation in Proppian function order. Wu et al. [14] carry out the largest computational comparative folklore analysis thus far, utilizing NLP techniques on 1,925 folktales spanning different cultures and showing that the moral value systems encoded in folk narratives are distributed according to cultural regions, thereby verifying that the structure and themes of folklore are culturally specific and identifiable via computation. Deka and Ojah [11] employ Propp's theory on Assamese folktales and find that there are systematic departures from the functional sequence of narratives as per the theory in the South Asian culture due to culturally unique narrative logic; hence, there is a case for using Proppian functions in diagnosis, not prescription. Billah et al. [9] offer a structural analysis of Bengali folklore through NLP techniques and prove that Bengali folk stories show unique genre characteristics that can be quantified via computation. The findings of these studies show that it is both practical and necessary to do structural computation in South Asian folklore; moreover, Proppian Inversion can be used for detecting westernized narrative structure in AI-generated tales.

2.3 Summary of Key Findings

The eight areas of thematic focus explored in the review above converge on three core findings, which collectively constitute the gap that forms the basis of the research proposed for this thesis. First, cultural erasure in LLM-generated text is a systematic and architectural feature of AI that is significantly more pronounced in low-resource cultural backgrounds, like Bengali folklore, compared to high-resource Western cultures. Second, there does not currently exist a reliable and validated evaluation framework for measuring cultural fidelity in AI-generated folkloric narratives. Finally, the human element of this problem; the question of whether culturally aware individuals can discern cultural erasure in AI text, under what circumstances, and how successfully, remains entirely unexplored.
Table 2.1 connects each thematic area to its central finding, and the gap that this research seeks to fill.

Table 2.1: Connection between thematic areas, key findings from the literature, and the gaps addressed by this study.

Thematic Area	**Key Finding from Literature**	**Gap Addressed by This Study**
LLM Cultural Bias	LLMs exhibit systematic Western default bias in value alignment, cultural reasoning, and language generation processes [20], [22], [30], [31].	No study evaluates AI cultural erasure in a folkloric retelling context with a validated multi-dimensional assessment model.
Low-Resource Bengali NLP	Bengali NLP systems underperform high-resource languages; Bengali dataset lacks cultural fidelity information [5], [9], [21], [24].	No evaluation framework or annotated corpus exists for assessing cultural fidelity in Bengali folkloric narratives generated by LLMs.
Evaluation Metric Adequacy	Standard NLP evaluation metrics (BLEU score, etc.) are inadequate for cultural evaluation; cultural-specific frameworks have been developed for high-resource languages [16], [27], [32].	No validated cultural fidelity evaluation framework exists for assessing folkloric traditions in non-dominant languages; the inadequacy problem has been established.
Folklore Corpus Construction	Translation pipelines, expert annotation, and cultural metadata construction are required for low-resource folklore corpus compilation [6], [7], [10], [13].	No corpus exists for evaluating Bengali folklore with multi-dimension cultural metadata for AI research.
HCI and Cultural Homogenization	AI writing assistants systemically homogenize the style of their output toward Western conventions; transparency tools are essential for culturally aware AI communication [17], [19], [23].	No study has investigated the HCI implications of AI-generated cultural homogenization for folkloric texts or assessed the risk of fluent deception.
Postcolonial Digital Humanities	AI systems perpetuate knowledge hierarchies established during the colonial era; sovereignty of representation belongs to communities; LLM knowledge of South Asian culture is documented as severely deficient [4], [25], [28].	No computational evaluation of cultural fidelity that incorporates a postcolonial perspective and utilizes authorized folkloric source material has been conducted.
Cultural Memory and AI Perception	Cross-cultural differences in memory specificity are moderated by age, cultural background, and familiarity with stories; the field of AI content detection is characterized by pervasive patterns of overconfidence [8], [26].	No empirical study has investigated the HCI implications of cultural erasure in LLM-generated folkloric texts.

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Thematic Area	**Key Finding from Literature**	**Gap Addressed by This Study**
Computational Folklore Analysis	Folklore exhibits structural differences in Proppian functions; Bengali folklore has identifiable structural elements; South Asian folkloric traditions diverge significantly from Western Proppian functions [9], [11], [14].	No computational framework has yet emerged for detecting Western function imposition in LLM-generated non-Western folkloric retellings.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

The research project was defined by these functional requirements that evolved over time through multiple iterations and were formalized before they were implemented. These functional requirements define what was included within the research, how well the work would be done, and when it had to be completed.

3.1.1 Functional Requirements

Table 3.1: Functional Requirements

Functional ID	**Description of Functional Requirement**	**Priority Level of Requirement**	**Description of Verification Procedure**
FR-01	Develop a reference corpus of 27 Bengali folktales collected from two culturally confirmed sources. Each collection will contain original Bengali texts, and each story is paired with a human translated English version. All versions are produced without normalization. The corpus is designed to reflect regional and dialectical variations.	High	Review of folkloric expertise; Audit of no-normalization policy; Agreement between annotators ($\kappa \geq .8$)
FR-02	Identify six large language models using four different prompt engineering techniques in two languages of interaction (English & Bengali). The large language model is used to generate 1296 unique versions of the folktales for systematized comparison.	High	Verify generation logs; Confirm number of artifacts generated ($6 \times 4 \times 27 \times 2 = 1296$)
FR-03	Test three potential designs for assessing the cultural accuracy of the folktales (design A — Lexical Fidelity Scorer; design B — structural alignment scorer based upon embedding; design C — holistic cultural auditor using a rubric) on the same corpus and use the results to make a data-driven decision regarding the best design.	High	Logs documenting execution; Artifact created during execution validated; Correlation coefficient for human alignment $r \geq .85$
FR-04	Use design C as the main mechanism for evaluating cultural accuracy and utilize a rubric-based method to provide a multi-dimensional assessment with hard logical constraints built into the rubric. Also document all intermediate scoring metrics in a structured JSON file format to allow for auditing and reproducing results.	High	Validate JSON output schema; Verify that logical constraints were adhered to; Validated correlation coefficient for human alignment
FR-05	Conduct a Human Verification study where participants are asked to determine whether or not a given retelling was generated via AI or authored by an expert across matched story pairs. Participants are also requested to provide confidence levels for their responses as well as written reasons explaining why they made those decisions. This provides additional insight into how effectively the automated system detects cultural distortions as well as assesses validity of automated assessments.	High	Obtain IRB approval; Document participant information ($n \geq 60$); Analyze statistical significance of relationship between human perceptions and automated alignments

3.1.2 Non-Functional Requirements

Table 3.2: Non-Functional Requirements

ID	Requirement Description	Priority	Verification Method
NFR-01	All datasets, evaluation logs, intermediate outputs and prompt templates shall be version controlled and remain in an auditable, unchanging state to allow all results from this project to be replicated independently.	High	Audit of the version control repository; confirmation that all pipeline elements using random seeds have documentation of their fixed random seeds.
NFR-02	Each of the three design approaches shall be implemented as separate evaluation modules and each module must execute without any dependencies on other modules or shared memory space and also must not cross-load any evaluation outputs.	High	Testing to confirm that the three design approaches can execute separately. Confirmation of unique file structures created during evaluation for each design approach.
NFR-03	All corpus annotations will be validated by at least one qualified expert in folklore who conducts a blind review. Additionally, all annotation schema will be formally locked down before any model evaluations commence.	High	Validation reports submitted by experts. Timestamp of when the annotation was locked down is confirmed to be before any model evaluation logs were generated.
NFR-04	The Human Verification Study shall comply with all applicable Institutional Review Board approved study protocols. No personally identifiable information shall be collected. Any participant data shall be anonymized prior to analysis. Data storage shall be done via secure and encrypted methods with limited access.	High	Approval from the Institutional Review Board. Audits to verify that personally identifiable information has been removed from participant data. Access to participant data storage logs.
NFR-05	Open source and free tier tools should be used wherever possible through all of the pipeline stages to reduce environmental impact while eliminating direct financial costs associated with running this pipeline. This should increase both the reproducibility and accessibility of this pipeline for future researchers.	Medium	Audit of the tool inventory to ensure only open source and free tier infrastructure has been used.

3.1.3 Research Constraints

Table 3.3: Research Constraints

ID	Limitation/Constraint Description	Implication for Interpretation and Scope of Results
C-01	Only pre-trained frozen black box language models are evaluated — no fine-tuning, retraining, or parameter modifying is ever performed at any stage within this research.	User experiences describe cultural erasure patterns in deployed LLM behavior; no corrective model-level interventions are assessed for this study.
C-02	The Reference Corpus includes two primary source collections: Bangladesher Lokoj Shongskiti Gronthomala (Bangladesh Academy); & Thakurmar Jhuli.	Bengali folklore as documented within these collections are represented by findings of this study; regional oral variation across entire country of Bangladesh is not captured.
C-03	Recruitment of human study participants uses exclusive networks and channels including social media platforms (BRAC University; Rajshahi University), community-based organizations and other groups that focus on culture.	Sample may not be representative of all demographic groups of Bengali speaking populations, especially rural elderly communities with limited access to digital technologies.
C-04	Study is conducted using only Bengali-language cultural content and evaluated in a bilingual context of Bengali and English.	Findings may not generalize to other non-Western or South Asian cultural traditions without further validation using culturally equivalent corpora.
C-05	All computational resources utilized for this study rely solely on free tier API access (Hugging Face, Open Router) and open source tools; dedicated proprietary compute resources were never provisioned.	Throughput of evaluation subject to shared infrastructure availability, rate limits set by free tier services; exact replication requires same level of free tier access when executing.

3.2 Societal Impact

3.2.1 Contributions to Cultural Preservation

To provide an initial assessment of the impact of LLMs on cultural knowledge and to support continued growth of the field, this study focuses specifically on two domains – cultural knowledge and the use of machine learning (ML) methods to preserve cultural knowledge. It uses an ethnographic case study of Bengali folktales as an example domain to explore how large language models systematically distort content from diverse cultural contexts.

The primary focus of the study is on assessing the extent to which LLMs alter or misrepresent content in Bengali folktales. A secondary goal is to assess whether such

changes can be identified using computational methods. To identify potential biases in LLM performance, we also assessed whether differences in query style, query intent, or model type result in different levels of distortion. Our results suggest that LLMs do systematically alter content in Bengali folktales; however, our analysis does not indicate consistent bias in either direction based on query characteristics.

We view this project as a pilot effort to begin exploring issues related to ML-based cultural knowledge preservation and to lay groundwork for future studies. We believe that developing methods to analyze and mitigate LLM-induced distortion of cultural knowledge will be essential to supporting equitable access to digital representations of diverse cultural heritages.

3.2.2 Societal Risks and Affected Communities

This article points out that there are many types of risks related to how society interacts with computers. Two of those risks identified in the study are "deceptive fluency" and "structural language inequality". Deceptive fluency occurs when AI uses words and grammar to create stories or messages that look like they were created by humans. However, instead of using local traditions and ways of thinking about things, AI replaces those concepts with ideas that are common in western cultures. This type of deception is not easy to identify because it doesn't generate an obvious error message, so people may believe what they read even though it is not accurate. The study confirmed that when AI has a high level of "cultural fidelity" (the degree to which an AI product is similar to its original culture) people cannot tell if the story was written by an AI or a human expert. In other words, the societal risk of reading false information is real and proven to be true.

Structural Language Inequality refers to the fact that in all five ways that the researchers evaluated the accuracy of the stories generated by AI – (the percentage of words retained from the original story; whether the story is grammatically correct; whether the story conveys an understanding of the original culture; whether the story conveys a sense of context; whether the story conveys a sense of time) – the AI produced better stories when they were told in English than when they were told in Bengali. In fact, in terms of retaining the original words in the story, Bengali stories contained 36% fewer words than the English versions. Additionally, two of the AI models failed completely when asked to produce Bengali versions of the stories. The researchers concluded that this was due to less money being spent on developing and training models in languages such as Bengali than on models in languages such as English. The authors point out that for individuals who speak only one language and use LLMs (language learning machines) primarily in their native tongue, they will receive a worse quality representation of their heritage through LLMs.

3.3 Environmental Impact

3.3.1 Computational Footprint and Mitigation

Inference-based only, the researchers did not train nor fine-tune a single new language model during the course of their investigation. As such, there will be no

additional environmental costs for the energy needed to train these large scale language models that would be included in the researcher's direct environmental costs. The environmental costs incurred by the researcher will be limited to the energy consumed by each inference (computational) call made to the language models as they generate and evaluate the various versions of their stories during the retelling generation and evaluation phase.

Each inference call made to the researchers' language models was made using free tier APIs provided by Hugging Face and Open Router. Both Hugging Face and Open Router offer free tier API services that allow users to make unlimited inference calls to their pre-trained language models. This means that the researchers could utilize an unlimited number of cloud servers without having to pay for them. Therefore, the researchers do not have to include the marginal environmental costs of utilizing additional hardware for this research. The researchers generated embeddings using Design B's paraphrasing model (paraphrase-multilingual-MiniLM-L12-v2) via Hugging Face's free tier inference. Design C utilized Gemini 2.5 Flash via its free tier API to evaluate each story. Each of the supplementary tools (Tesseract OCR, The Fuzz, Python, etc.) used by the researchers were open source and did not require any cloud resources.

The researchers evaluated 1296 different narrative artifacts based upon the three different designs, requiring approximately 46 hours of computing time in total. The majority of the computing time was allocated to Design C since Design C had multiple structural language model inference calls in order to support the evaluation process. These results indicate that the researchers have produced a low-to-moderate level of environmental footprint given the nature of the research being done; especially when compared to other studies involving training large scale language models and/or deploying those models into production environments. By purposefully selecting to utilize shared and free tier infrastructure in every phase of the research, the researchers have managed to minimize their environmental impact by avoiding the need for dedicated resources and producing a research pipeline with an overall significantly smaller environmental footprint than previous similar research that has been performed on provisioned hardware.

3.4 Ethical Issues

3.4.1 Human Subject Research Ethics

This study was reviewed and cleared by the Institutional Review Board of BRAC University before collecting any data. Prior to interacting with any participant materials, each member of the research team completed the Collaborative Institutional Training Initiative (CITI) certification course on ethics related to working with human subjects.

Prior to participating, each potential participant was explicitly notified that they could withdraw from this study at any time and that there would be no negative academic, institutional or social consequences. In addition, the participants were explicitly told at the beginning of the study that AI generated content would be

present in each of the standard story sets. All participants were also required to give their full consent before proceeding to the actual test. All participants received a full debriefing after completing the study where they were told what the purpose of the study was and offered the option to remove their responses from the study at no cost or penalty. No personally identifiable information was ever collected at any point in time in this study. All participant responses were anonymized prior to analysis. Data collected in this study were stored in an encrypted format so that access to those data was restricted solely to members of the research team who were authorized to do so. Some demographic data such as age range, education history, language proficiency level in Bengali, frequency of use of AI tools and degree of familiarity with folkloric stories were collected at the categorical level and are unable to be associated with specific individuals.

3.4.2 Cultural Ethics and Community Ownership

In addition to simply preserving them as historical records, folklore provides an opportunity for people to express their culture and identity. For researchers using these types of stories as sources of information, there are additional commitments beyond those typically found in academic research ethics.

In order to consider whether the use of *Bangladesher Lokoj Shongskiti Gronthomala* would be legitimate as a source of information and if there would be a need for community approval prior to conducting research using the collection, the lead researcher visited the Bangla Academy in Dhaka to obtain formal permission from the organization to utilize the *Bangladesher Lokoj Shongskiti Gronthomala* as a resource for research. The method of obtaining formal permission represents an acknowledgment of community control over their own cultural resources consistent with values expressed within the literature related to post-colonial digital humanities (Risam, 2019), and the guidelines established by the United Nations Special Rapporteur on Cultural Rights for ethically responsible practices when digitizing cultural heritage (Xanthaki, 2024).

Thakurmar Jhuli was accessed under its public domain designation.

A “no normalization” methodology was used during the construction of the corpora for all three databases to preserve the natural dialectical variations present in the language in lieu of standardizing the texts into some common, often more prestigious literary form, thereby acknowledging and respecting the internal variability of Bengali folklore.

Both the Reference Corpus and the annotation associated therewith have been clearly defined as one possible interpretation of each selected narrative, but are not intended to represent any type of universal or definitive cultural norm. All participants who served as expert validators of the narratives were encouraged to make decisions based upon their individual scholarly judgments and not according to some pre-defined evaluation template; further, any inter-annotator discrepancies were resolved through discussion amongst the evaluators rather than through hierarchical means.

One significant ethical consideration associated with this project relates to the ambiguities associated with authorship of AI generated cultural content. As AI generated versions of traditional narratives become increasingly sophisticated and similar in structure and syntax to those authored by humans, there exists a growing concern that they may be misinterpreted as original creative products developed and authored by members of local cultures. Not only does this risk exist academically, but also in terms of how cultural content is memorialized and archived online. This project has taken the first steps toward identifying this potential risk, and documenting where/when human-based detection failed to identify such risks; however, this project identifies developing design strategies to provide transparent authorship for AI writing systems as critical for future research.

Finally, while this project recognizes that achieving full participatory co-design with originating communities — in which community members collectively establish and define what aspects of their cultural traditions cannot be replaced — will require significantly more time and effort than can currently be provided within the context of this current project. Therefore, participating in community-driven efforts to develop shared understanding and consensus regarding culturally replaceable versus non-replaceable elements of their collective cultural heritages will be recognized as an area of focus for future research projects.

3.4.3 Legal and Compliance Considerations

In order to utilize published folkloric sources in the research,a couple things had to be taken into consideration. All the data taken from *Bangladesher Lokoj Shongskiti Gronthomala* was taken after obtaining an official academic research license in person at the Bangla Academy. Data from *Thakurmar Jhuli* was utilized due to its public domain status. The research did not reproduce nor commercially sell/distribute the actual source materials used in this study; rather the source materials served as references, only to compare/evaluate them to the AI generated retellings. None of the AI generated outputs produced in this study were ever presented as legitimate folkloric content, publicly displayed or sold/distributed without disclosing that they were produced by an artificial intelligence system.

The Language Model Inference was conducted under the terms of service of each of the free-tier API providers that provide open access language models. Open Router and Hugging Face's free tier usage for academic/non-profit purposes fall well within the permissible uses defined in each of their respective service agreement. The Human Verification Study complied with all relevant applicable data protection requirements. There was never any collection of sensitive personal data; there were no transfers of data out of the research group; and none of the participant data was retained longer than needed to complete analysis and verify results.

3.5 Standards

The above research is subject to the below guidelines. Each guideline was followed during the planning, implementation, and reportage of the research process described in this document.

Table 3.4: Applicable Standards & Codes of Practice

Standard/Code	**Issuing Organization**	**Applicability to This Thesis**
IEEE Code of Ethics	Institute of Electrical and Electronic Engineers	Defines how researchers should accurately describe methodologies and results used in their research and avoid making biased claims about technology they are using.
ACM Code of Ethics & Professional Conduct	Association for Computing Machinery	Defines the ethics of designing and implementing an evaluation tool for artificial intelligence systems that evaluates participants fairly, safely protects participant information collected through the research and ethically reports research findings on the performance of artificial intelligence systems.
APA Ethical Principles of Psychologists and Code of Conduct – Human Research Guidelines	American Psychological Association	Specifies the design for the Human Verification Study, which includes how informed consent will be acquired, the IRB-approved mild deception/debriefing procedure, safeguards for participant well-being and confidentiality requirements for participant data collected during this study.
BRAC University Institutional Review Board (IRB) Approval Process	BRAC University	Before collecting data, all participant-related research procedures associated with this project were reviewed and approved by the IRB at BRAC University before conducting the data collection portion of the project.
Terms of Service, Open Router API	OpenRouter Inc.	Calls to the API for retelling generation were made within the permissible scope of non-commercial academic research provided in the OpenRouter free-tier service agreement.
Terms of Service, Hugging Face	HuggingFace Inc.	Generating embeddings and inferring models were performed consistent with HuggingFace's permitted uses for academic and research purposes.
Research Use Permission from Bangla Academy, Government of Bangladesh	Bangla Academy, Government of Bangladesh	Academic research permission was formally granted in-person to Bangla Academy to allow for its use as the primary corpus resource in this thesis.

3.6 Project Management Plan

3.6.1 Project Overview and Phases

A multi-Phase approach was chosen to help ensure a systematic process to develop and evaluate a tool designed to automate an important but time-consuming task (i.e., creating culturally faithful adaptations). Three discrete phases were planned, with well-defined goals and deliverables for each Phase. Each Phase is dependent on the completion of the previous Phase. In other words, the Reference Corpus must be built before automated evaluations can begin; the automated evaluations must occur before the stimuli for the Human Verification Study can be selected; etc. Therefore, this sequence allows for quality checks at the end of each Phase. That is, if some problem or weakness is found during one Phase, it can be corrected before proceeding to the next Phase. Table 3.5 shows the project activity timeline and Figure 3.1 represents the Gantt chart for this project.

Table 3.5: Project Activity Timeline

Phase	**Activity**	**Period**	**Key Milestones**
Pre-research	Thesis topic ideation & research question definition	Mid-July 2025	Research questions and scope defined
Phase 1	Literature review across eight thematic areas	Early Aug – Early Oct 2025	30 papers reviewed / theoretical framework identified
Phase 1	Identify source materials; obtain in person permission from Bangla Academy	Early Oct – Mid Oct 2025	Select folktales; institutional permission granted
Phase 2	Digitize the Reference Corpus using OCR technology & correct manually; translate benchmarking & human post editing	Mid Oct – Early Dec 2025	Reference corpus digitized & translated (MQM 91.2)
Phase 2	Determine which language models are suitable; determine how to create prompts & generate 1,296 AI retellings	Mid Nov – Mid Dec 2025	1,296 narrative artifacts generated & logged
Phase 2	Develop architecture of cultural fidelity perception framework & specify dimensions of evaluation	Mid Dec – Late Dec 2025	Architectural design of framework & four dimensions completed
Phase 2	Implement designs A, B & C as separate evaluation approaches	Early Jan 2026	All three design methods operational & testable
Phase 2	Annotate 54 story artifacts by experts across four dimensions of cultural fidelity	Early Jan – Early Feb 2026	All annotated files structured & saved as JSON
Phase 2	Compute inter-annotator reliability; blind expert validation of annotations; lock annotation schema	Early Feb 2026	$\kappa = .893$ achieved; annotations locked
Phase 2	Pilot evaluation: evaluate 10 stories using all three design methods	Early Feb – Mid Feb 2026	Compare design methodologies; selected Design C as primary evaluator
Phase 2	Complete automated evaluation of all 27 stories under Design C; scoring results dataset available	March 2026	1296 retellings evaluated; results dataset complete
Phase 3	Develop protocol for Human Verification Study; determine which story stimuli will be used based on Design C scores	Early April 2026	Study protocol finalized; IRB scope confirmed for probe condition
Phase 3	Develop survey instrument (using Google Forms); create pair of stories/probe conditions for participants to respond to	Mid-April 2026	Survey ready for deployment
Phase 3	Analyze quantitative data from Design C's automated evaluation; conduct qualitative analysis of automated evaluation findings	Late-April 2026	Automated evaluation findings complete
Phase 3	Deploy Human Verification Study (Part 1: high-fidelity story pairs / Part 2: low-fidelity story pairs); collect responses from 76 participants	May 2027	Participant responses collected
Phase 3	Quantitatively and qualitatively analyze responses to human study questionnaire; perform alignment between human-automated evaluations	Late-May – Early-Jun 2027	Full results analysis complete

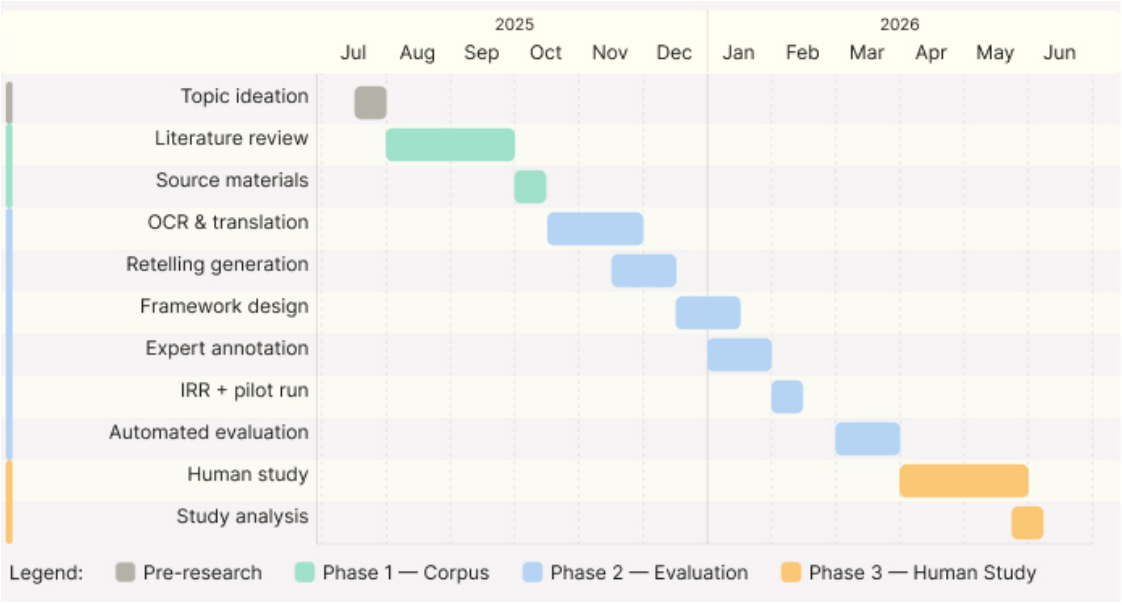


Figure 3.1: Gantt Chart

3.6.2 Quality Assurance

Quality was consistently preserved in this study via four quality control measures that were built directly into the design of the study. First, all of the corpora were reviewed independently from one another (blind reviews) by two folklorists, and they both identified entries that required revision as a result of their respective reviews; revisions continued until consensus was reached on an entry for each pair of reviewers. The results of this process resulted in the development of an independent annotation schema which was then frozen prior to evaluation commencing.

Secondly, an Inter-Annotator Agreement (IAA) analysis was completed for all 54 narrative pairs, resulting in a Cohen's Weighted Kappa statistic value of $\kappa = 0.893$ and a raw agreement rate of 96%, providing evidence that the annotation schema was reliable prior to beginning evaluations. Thirdly, all of the datasets utilized during the course of this study, along with the evaluation scripts, prompt templates, and output artifacts generated by the models evaluated, were stored using a version-control system. Additionally, all of the stochastic elements within the pipelines were run using fixed random seeds to enable deterministic reproduction of the same outputs when running different iterations of the pipelines used to evaluate the models. Finally, the annotation schema was locked prior to evaluating any of the models, thus preventing any possible modifications to the evaluation criteria after reviewing any model outputs.

3.7 Risk Management

This research project is characterized by its interdisciplinarity in terms of natural language processing, computational folklore analysis, and human-computer interaction. Therefore it has risks on four levels: technical, methodological, human participant, and ethically. In order to proactively manage these risks a risk management

strategy was therefore used early on during the planning phase of this project in order to identify the risks associated with the implementation of this project prior to the start-up of the research activities and to integrate measures of mitigation within the research design itself. The complete Risk Register can be seen in Table 3.6.

Table 3.6: Risk Register

ID	Category	Risk Description	Likelihood	Impact	Severity	Mitigation Strategy	Residual Risk
R-01	Technical	Bengali OCR may produce incomplete or damaged text due to conjunct consonants, dialectal spelling conventions and historical writing scripts.	High	High	Critical	All OCR results will be manually checked line-by-line against printed originals; no automatic normalizations allowed.	Low
R-02	Technical	Machine translation systems tend to "flatten" culturally-specific terms, resulting in a Reference Corpus that does not reflect the culture-specific context of the original texts.	High	High	Critical	Multiple MT systems compared using MQM methodology; Gemini 2.0 Flash chosen for drafts; all drafts underwent manual post-editing to create a cultural "air gap" before the Reference Corpus was finalized.	Low
R-03	Technical	Interruptions or failures in free tier API operations (downtime, rate limits or safety filters) may disrupt the evaluation pipelines.	Medium	Medium	Moderate	Automatic retries if necessary; individual log entries per item; modular pipeline design ensures continuation even if one module fails.	Low
R-04	Technical	Model results are either incoherent (e.g., garbled script) or empty and cannot be evaluated under standardized criteria.	Medium	Medium	Moderate	Script generation failures categorized as a new failure type; incoherent results identified and documented; no standard scores assigned.	Low
R-05	Methodological	Annotator judgments regarding cultural interpretation may result in inconsistencies among different annotators.	Medium	High	High	Cultural annotation guidelines formalized before expert engagement; two independent folklore scholars validate annotations blindly; IRR analysis performed; schema locked after revisions.	Low
R-06	Methodological	Embedding-based evaluations (Design B) may exaggerate cultural fidelity by confounding semantic similarity with cultural authenticity.	High	High	Critical	Three independent designs implemented; Design B failure cases documented empirically; Design C (hard logical constraint override) designated as main evaluator after comparison.	Low
R-07	Methodological	Unclear boundaries across multiple domains (NLP, HCI, folklore studies, AI evaluation).	Medium	Medium	Moderate	Strict adherence to project phase plan; objectives formally stated at outset; milestone-based supervisory structure; model-level interventions deliberately excluded from scope.	Low

ID	Category	Risk Description	Likelihood	Impact	Severity	Mitigation Strategy	Residual Risk
R-08	Human Participant	Underrecruitment of participants in target age groups and familiarity subgroups.	Medium	Medium	Moderate	Multi-channel outreach via BRAC University and Rajshahi University networks, local cultural organizations and social media; both study parts can run independently with separate samples.	Low
R-09	Human Participant	Prior awareness of AI-generated material would cause demand characteristic effects, invalidating naturalistic cultural perception data.	High	High	Critical	Mild deception protocol approved by IRB; participants unaware of AI generation before study; full debriefing immediately after all tasks; option to opt-out of data inclusion provided.	Low
R-10	Ethical/ Cultural	Evaluation framework assumes a singular view of cultural authenticity, potentially misrepresenting the internal diversity and evolving nature of Bengali folkloric traditions.	Medium	High	High	Fidelity assessed relative to expert-validated reference point only, not as universal authenticity; cultural humility integrated into research philosophy; limitations discussed in Section 5.6.	Medium
R-11	Ethical/ Cultural	Unauthorized access to study response records resulting in a breach of participant data privacy.	Low	High	Moderate	Only non-personally identifiable information collected; all data anonymized prior to analysis; encrypted storage with access restricted to authorized personnel only.	—

3.8 Economic Analysis

3.8.1 Overview

Although the purpose of this project is for educational purposes, it still had to be deliberately managed as a means of allocating and managing the different forms of resources (labor, hardware/software, and data) necessary to complete the project. The fiscal profile of this research differs from other types of research in that the total amount of money spent on direct expenses was equal to $0.00 due to the fact that: I intentionally selected tools that would provide me with the opportunity to perform all language model computations and embeddings for free via free tier API access on shared cloud services; All tools utilized in the pipeline are open-source; No compensation was given to any of my participants; Source materials were accessed at no cost by obtaining formal permission from an institution. Therefore, the only resource which I invested significant amounts of time into and which represented the majority of the project's expense were skilled human labor.

3.8.2 Direct Costs

This study received all direct access to computational resources at a $0 dollar price point. Language model inference for the generation of 1296 AI retellings utilizing the open router free-tiers was conducted with Meta Llama-3.3-70B-Instruct, GPT-OSS-120B, Qwen-2.5-7B-Instruct, DeepSeek-R1-0528, Gemini-2.0-Flash-Exp,

and Mistral Devstral-2512. In addition to language model inference for Design B's embedding generation, we utilized paraphrase-multilingual-MiniLM-L12-v2 via the free-inference-tier on HuggingFace. Design C evaluations used Gemini 2.5 Flash as part of the free-tier API. OCR processing was done using Tesseract (an open source engine). All supporting libraries—TheFuzz, Sentence-Transformers, and Python Scientific Computing Ecosystem—are open source. No participant incentives were given; all participants in the Human Verification Study participated voluntarily. The Institutional Permission from the Bangla Academy, which is responsible for providing us access to our primary source collection of folklore, was obtained at no cost after visiting their offices in person. Thakurmar Jhuli is a work that has entered into the Public Domain.

Therefore, the total dollar amount spent in accessing necessary resources directly related to this research project is $0.

3.8.3 Human Resource Investment

The one true "substantive" resource investment made by the project was the time and resources in terms of people skills of those that were part of the research team. A breakdown of estimated hours worked per research task is shown below in Table 3.7; these are a general representation of the amount of time required to complete each research task and to be able to demonstrate the "labor intensity" of each task as it relates to the entire project timeline.

3.8.4 Cost-Benefit Analysis

As of July 2026, the research produced the following long-term assets against a total direct monetary expenditure of zero dollars. A validated Bengali folklore corpus consisting of 27 stories with structured cultural annotations along the four dimensions of culture was developed. The preservation policy adopted by the research project was to preserve all artifacts in their original format without normalization. An annotation schema was developed which can be reused in future studies of AI generated culturally diverse content. A validated framework for evaluating the Cultural Fidelity Perceptions (CFP) of culturally relevant AI applications was developed and it is adaptable to other low resource cultural traditions. A dataset of 1296 fully evaluated AI generated retellings of the same stories as those within the folklore corpus were developed. Each artifact in the dataset contains per-artifact scoring records across three independent design approaches. Finally, an experimentally tested human-centered methodology for evaluating AI systems based on the perceptions of culturally relevant applications has been developed. The methodology provides a replicable experimental design for HCI and NLP researchers. In addition to demonstrating these findings, the research demonstrated a new finding of independent methodological significance: rigorous, multi-component cultural AI evaluation at scale is possible at nearly zero direct monetary cost when using strategic combinations of open source and free tier cloud computing infrastructure.

3.8.5 Scalability and Sustainability

The main limitation of the scale-up of this project is based upon human annotators rather than computation. An extension of the proposed framework to include larger corpora, different cultural traditions, or larger samples will have to expand both the amount of work required by annotators and experts for validating results as well as the number of people involved with coordinating studies. However, these expansions do not represent direct dollar expenditures at present. Additionally, the fact that the entire architecture of the system utilizes free and/or open-source software ensures that the costs associated with each step can be continued into the future through an economical process while eliminating the need for expensive licensing fees, cloud computing, or institutional computer allocations.

Table 3.7: Resource & Costs Summary

Category / Activity	**Description**	**Infrastructure Model**	**Approximate Cost**
Direct Monetary Costs			
Inference — Retelling Gen.	1,296 retellings, using 6 models & 4 prompts	Free Tier Open Router API	$0.00
Inference — Cultural Aud. (Design C)	1,296 structured evaluations of LLMs	Free-Tier Gemini 2.5 Flash API	$0.00
Inference — Translated Draft (Corpus)	27 translations of source texts (Gemini 2.0 Flash)	Free Tier API	$0.00
Generation Embeddings — Design B	1,296 narrative embeddings (MiniLM-L12-V2)	Free Tier Hugging Face	$0.00
Licensing Fees For Software	Python, TheFuzz, Sentence-Transformers, Tesseract, All Dependencies	Open-Sourced	$0.00
Source Material — Banglader Lokoj Sankshipti Grantha	17 Oral Folktales from six regional linguistic zones	Formal Bangla Academy Permission (In-Person)	$0.00
Source Material — Thakurmar Jhuli	10 Literary Folk Tales Selected From Collection	Public Domain	$0.00
Compensation for Participants	76 Volunteers	Human Verification Study participants are volunteers	$0.00
Direct Total Dollar Costs for Project			**$0.00**
Human Labor			
Labor — Literature Review	30 Papers; Critical Reading and Synthesis of Surveys	Research Labor	120 Hours
Labor — Corpus Construction	Manual Correction, OCR Executions, Post-Editing of Translations	Skilled Research Labor	180 Hours

Category / Activity	**Description**	**Infrastructure Model**	**Approximate Cost**
Labor — Annotation and Validation	Schema Development; Expert Coordination; Creation of 54 JSON Annotations; IRR Analysis	Skilled Research Labor	120 Hours
Labor — Framework Development	Implementation of Designs A, B, C; Creating Evaluation Pipeline	Skilled Research Labor	100 Hours
Reviewing Evaluation Output	Evaluation Pipeline executed 1,296 times; Failure Modes reviewed; Output reviewed	Skilled Research Labor	60 Hours
Labor — Human Study	Protocol design; Instrument development; Deployment; Data Management	Skilled Research Labor	80 Hours
Labor — Analysis & Manuscript	Statistical Analysis; Qualitative Coding; Manuscript Writing/Preparation	Skilled Research Labor	180 Hours
Total Labor Estimates for Project Personnel			**840 Person-Hours**

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

4.1.1 Design Philosophy

The methodological process employed in this research has been guided by four core principles that are applicable to each and every design choice made during the development of this project.

Firstly, there is diagnostic intent; the purpose of this system is to automatically recognize, assess quantitatively, and define qualitatively, how patterns of cultural erasure emerge in the output from an artificial intelligence (AI) generated text-based response. The intention of the system is not to correct responses, nor to assess the performance of an AI model generally.

Secondly, the automated evaluation mechanism will be used as a measurement device for validating human cultural judgments, rather than being a replacement for them.

Thirdly, all evaluation mechanisms generate structured intermediate data products that can be examined, contested, and replicated independently with regard to proprietary internal information.

Fourthly, cultural humility; there is no single or authoritative component of the framework that defines what constitutes cultural authenticity. Each scoring assessment made within the framework is based on one expert-defined reference interpretation of each source narrative, and the framework recognizes that Bengali folklore has greater diversity than this reference.

4.1.2 Theoretical Framework

Vladimir Propp's morphological analysis of folk narratives indicated that 31 narrative functions recur relatively constantly in a sequential order in Russian folktale (Propp, 1928/68). Cross cultural applications demonstrated that the Proppian functions are not universal but represent culturally specific forms of logic for the construction of a story, with different traditions exhibiting different functional se-

quences, missing function and combinatorial patterns (Deka & Ojah, 2022; Wu et al., 2023).

This study applies what is known as the Proppian Inversion to diagnose rather than prescribe use of Proppian functions to identify what Western conventional structures an artificial intelligence system has injected into a retold version of a story through imposition which were absent from the original. A structural displacement indicator (SDI) was defined as the count of Proppian functions present in an AI retelling that do not appear in expert-annotated reference corpus for that story. High SDI values indicate that an artificial intelligence system has introduced alien structure(s) onto a narrative tradition that does not employ these structures.

The Proppian Inversion converts a diagnostic template for a prescriptive structural template into a diagnostic contamination detector. It is claimed that while some Proppian functions may be absent from Bengali folk tales, any Proppian functions that exist in AI retellings and are absent from the reference corpus are evidence of displaced structural elements.

4.1.3 Four Evaluation Dimensions

Each dimension targets a distinct layer of cultural loss, a claim grounded in convergent findings across translation theory, folklore studies, and cognitive narrative theory, and together they provide full-spectrum coverage of how cultural meaning degrades in narrative retelling.

As a basis for the Lexical Integrity criterion, prior research in culture-bound item theory reveals that named objects, cultural terms, cosmological figures, and locations in source material are conceptually irreplaceable; substituting them with generic target culture versions entails a form of semantic loss that is ontologically different from paraphrase, as the object referred to and its cultural significance are inextricably linked [1], [15]. For the Narrative Fidelity criterion, it is established that schematization — the organization of story events into a causal sequence — is distinct from the surface level contents of the narrative, both cognitively and culturally; Yang et al.[33], for example, reveal through fMRI that schematic transmission and paraphrastic transmission draw on independent neurocognitive pathways, while Peritos [34] finds through empirical observation that younger storytellers consistently eliminate structural elements of a plot before they delete lexical elements, suggesting that fidelity to the narrative structure decays in a manner independent from lexical fidelity. Finally, the Socio-Cultural Marker Preservation criterion is supported by adaptation literature: in analyzing an indigenous film adaptation, Fatzinger [2] concludes that socio-cultural content such as social role relationships, kinship relationships, and culturally specific conflict resolution mechanisms cannot be reduced without homogenizing the story's core meaning.

Thus, the cultural authenticity of every single AI-generated retelling was evaluated using the four different dimensions, each representing one type of cultural loss (surface, structural, semantic, etc.):

Lexical Integrity represents if culturally-specific Bengali words are present, altered to English, or removed from an AI-generated retelling. Lexical integrity specifically evaluates the surface aspect of culture.

Narrative Fidelity represents if the story's structure (plot), order of events, and causality have been preserved when comparing the source story with the AI-generated version. Narrative Fidelity represents the structural aspects of a culture that exist independently of lexical surface representations.

Socio-Cultural Marker Preservation evaluates if culturally-specific practices (rituals, social roles, values, cosmology, relationships) are represented in an AI-generated retelling or replaced with general equivalencies. Socio-Cultural marker preservation represents the semantics of a culture's representation.

Structural Displacement Indicator is based on the Proppian Inversion and represents the total number of western-style narrative functions inserted into an AI-generated retelling. A Structural Displacement Index of Zero means there is no structuring influence from Western style storytelling applied to a non-Western tradition.

4.1.4 System Architecture Overview

The Evaluation Architecture has two Layers. The first layer is called the Evidence Preparation Layer; this layer accepts both the reference corpus annotations and the AI generated retelling of those references, and then converts that data into an "evaluation ready" format in which all the data for each artifact can be inputted directly into one or more of the various components of the Scoring Layer. Each of the three candidate design approaches (Design A, Design B & Design C) will be receiving the exact same prepared evidence, and will generate their own independent score for each artifact.

4.2 Preliminary Design or Design (Model) Specification

Three alternative independent design methods for addressing the cultural fidelity measurement issue have been developed. Each method captures one unique measurement paradigm, uses one unique technical methodology and has produced one unique measured output. Rather than commit to one method, three alternatives were developed so that an empirically-based evaluation could occur as to which method best measures cultural alignment. A comparison will be performed on the selected method to see if it demonstrates correlation with human based cultural judgments; this selected method will then be designated as the primary evaluator. The results of the comparative study used for evaluating the relative merits of the candidate designs are discussed in Chapter 5.

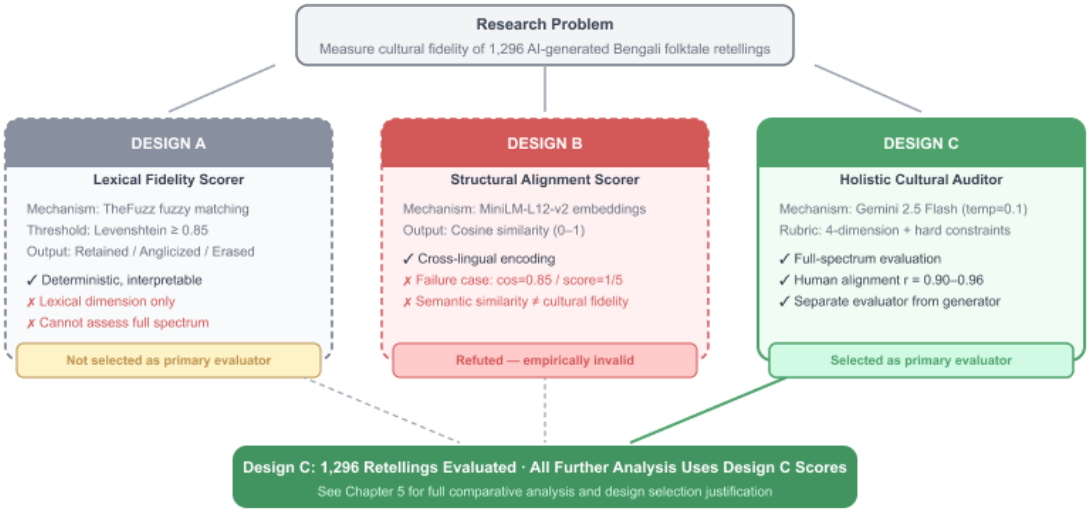


Figure 4.1: Three Candidate Design Approaches for Cultural Fidelity Evaluation

4.2.1 Design A: Lexical Fidelity

Measurement Philosophy: While the irreducible, unique specification of named entities in folklore — supernatural beings; locations; rites and practices; and relational titles carrying unique cultural meanings that are untranslatable into generic English — contributes to the cultural authenticity of those stories, Design A's measurement exclusively addresses the extent to which culturally registered Bengali words/phrases from the reference corpus retain or replace their original forms in the AI rendition.

Mechanism: In order to apply Design A to measure how well an AI rendition retains culturally registered Bengali terms from a reference corpus, each AI rendition is compared to the `lexical_terms[]` annotation block of the corresponding reference story. For each Bengali word/phrase annotated in the reference story, a Levenshtein-based fuzzy matching algorithm using TheFuzz library is applied to find the closest string match in the AI rendition. If the match has a normalized value greater than 0.85, then the term is considered retained and the AI rendition preserved the exact term. A normalized match score range of 0.50–0.84 indicates anglicization, where there was modification of the original term to preserve some degree of similarity through phonetics/morphology. If the normalized match score is less than 0.50, then the term is considered erased, as there is no longer a link between the term in the reference story and the AI rendition, due to replacement by either a generic or culturally unspecified term. The lexical retention rate (LRR) for each rendition is determined by dividing the number of terms annotated as retained and divided by the number of terms annotated in the reference story.

Limitation identified: Design A is capable of identifying only one aspect of cultural fidelity to a very precise degree using a completely deterministic approach. The main limitation of design A is its inability to measure narrative fidelity, sociocultural marker preservation, and structural displacement. A tale which systematically removes all plot-level aspects of cultural elements while preserving the surface

Table 4.1: Component A Mean Lexical Retention Rate by model and prompt strategy for Bangla Retellings

Model	Prompt 1	Prompt 2	Prompt 3	Prompt 4
MistralAI	0.038	0.078	0.025	0.513
OpenAI	0.064	0.067	0.053	0.440
Gemini	0.064	0.052	0.011	0.390
Deepseek	0.083	0.078	0.031	0.288
Llama	0.011	0.033	0.000	0.097
Qwen	0.011	0.000	0.000	0.051

Table 4.2: Component A Mean Lexical Retention Rate by model and prompt strategy for English Retellings

Model	Prompt 1	Prompt 2	Prompt 3	Prompt 4
Deepseek	0.102	0.192	0.082	0.681
OpenAI	0.082	0.187	0.057	0.665
MistralAI	0.102	0.123	0.077	0.567
Llama	0.119	0.123	0.122	0.547
Qwen	0.102	0.114	0.082	0.586
Gemini	0.093	0.159	0.102	0.432

level of Bengali terms will be considered a high-LRR score story even though it's highly distorted in terms of culture.

4.2.2 Design B: Structural Alignment

Measurement Philosophy: Cultural Fidelity and Narrative Structure: In order for the relational structure of an event-based plot to be culturally authentic within the context of a narrative, it must preserve the relationships among the structural elements of the events in the story. These include: (a) the temporal relationship of events; (b) causal relationship(s); and (c) logical relationships among events that demonstrate change or transformation.

Design B provides measurement of the similarities in the semantic representation of the narrative arcs of the AI-generated version of the story versus its reference narrative version based on the premise that there will be a positive correlation between structural and thematic similarities and cultural fidelity.

Mechanisms: To measure the degree of similarity between the semantic representations of each of the two narratives, both narratives were transformed into dense vector representations by applying a pre-trained multi-lingual model called `paraphrase-multilingual-MiniLM-L12-v2`, which can create cross-lingually consistent representations of sentences that can represent inputs in either Bengali or English. Then, using these vector representations, we calculated the cosine similarity (cos-sim) between them. The cos-sim values fall in the continuous range $[0, 1]$ and provide a measure of how close the two vector representations are to one another.

The closer the value of cos-sim is to 1.00, the closer the semantic representations of the two stories are.

Limitation/Identified Failure Case: The limitation/failure case of Design B was recognized through testing, and is shown in Table 4.3. Through testing, an example was found of an AI generated version of a story that achieved a cosine similarity score of 0.85 compared to its reference version but had replaced all entities/practices/cosmologies in the original story that are specific to culture with generic Western counterparts. Because the same abstract narrative structure (e.g., "the hero meets an antagonist," "the hero seeks supernatural help to overcome an adversary," etc.) had been preserved across versions of the story but because all cultural content had been eliminated, the test demonstrated that Design B does not accurately identify instances where cultural fidelity exists when the narrative structure has been preserved but all cultural content has been removed.

Table 4.3: Component B Failure Case — High Cosine Similarity with Complete Cultural Replacement

Narrative Element	**Reference Corpus Annotation**	**AI Retelling Content**	**Cultural Status**
Supernatural antagonist	*Rakkhosh* (Bengali ogre-demon)	"A powerful dark sorcerer"	Erased
Protagonist's spirit guide	*Bastu Devi* (household deity)	"A mysterious supernatural helper"	Erased
Sacred river crossing	Padma River (with ritual significance)	"A wide, dark river"	Erased
Resolution mechanism	Lord Shiva's divine intervention	"Ancient magic and sacrifice"	Erased
Cosmological framing	Three-world structure (*Swarga/Mrityulok/Patallok*)	"The spirit realm"	Erased
Kinship title of protagonist	*Bou* (daughter-in-law, carrying specific cultural obligation)	"The young woman"	Erased
Component B Cosine Similarity			**0.85**
Component C Mean Score			**1 / 5**

This failure mode occurs due to a property of sentence embedding spaces: semantically identical plots represented in culturally distinct ways result in similar vector representations. Thus, when a sentence embedding space is primarily trained using English language data, then this space represents abstract narrative schemas (not their cultural instantiations). Therefore, Design B systematically overestimates cultural fidelity whenever a narrative structure is preserved but when cultural content is completely replaced — which is exactly what this study is focused on.

4.2.3 Design C: Holistic Cultural Assessment

Measurement Philosophy: Cultural fidelity does not merely relate to how well a culturally-authentic retelling covers the lexicon and/or structural components of the original text. In order to maintain cultural authenticity, all of these aspects must be preserved together — including but not limited to, the original's naming conventions, relational structures, cosmological references, and narrative morality mechanisms. Design C embodies this comprehensive perspective through a structured rubric-based assessment provided by a large language model judge under explicitly-defined cultural constraints.

Mechanism: Each culturally-specific retelling produced by the AI will be judged using Gemini 2.5 Flash at temperature 0.1 — a very-low temperature setting which minimizes stochastic variation in rubric application while maximizing scoring reliability. The judge will use a structured scoring rubric to assess each culturally-specific retelling across the four fidelity dimensions previously defined in Section 4.1.3, where each dimension will receive a normalised 0-1 rating. Importantly, Gemini 2.5 Flash will function solely as the automated judge within this investigation; the culturally-specific retellings created by the Gemini 2.0 Flash model, representing a separate model from a previous generation, will not influence the results of this evaluation since both systems have separate training histories.

Following the rubric-scoring process, hard logical constraint override rules will be automatically applied to prevent potential overestimation in cases where the rubric-scoring may reflect surface-level plausibility rather than true cultural fidelity. The rules for these constraint logic overrides are described in Table 4.4. These rules represent an iterative design improvement based upon analysis of preliminary evaluation data collected prior to the commencement of the final evaluation.

Table 4.4: Hard Constraints Implemented by the Holistically-Based Cultural Auditor

Constraint ID	**Trigger Conditions**	**Action Taken**
HC-01	All annotated Bengali cultural terms identified as "Erased" (i.e., LRR = 0)	Set Mean Score Ceiling to 0.1 Regardless of Rubric Output
HC-02	Name of primary protagonist has been replaced by a non-Bengali or non-South Asian term	Set Mean Score Ceiling to 0.4
HC-03	Location(s) in which the central story occurs has/have been moved to a non-South Asian geographical location	Set Mean Score Ceiling to 0.4
HC-04	Proppian function that was not present in the reference material	Increment SDI by 1; Set Mean Score Ceiling to 0.7
HC-05	Retelling generates either empty or nearly-empty response (Script Generation Failure)	Set all component scores equal to 0; flag failure mode separately

Validation of Human Alignment: Before running Design C on the complete evaluation dataset, it was tested for accuracy in comparison to sets of human reference evaluations. A sample of forty stories drawn from the pilot dataset was assessed for a subset of the same four dimensions of fidelity by two researchers on the research team who had no access to any information regarding the ratings from any of the three designs. The Pearson product-moment correlation coefficient values comparing Design C-generated scores to those of human evaluators are listed in Table 4.5. For all four evaluation dimensions, there exist correlations exceeding 0.9 between Design C-scores and those from human reference evaluations demonstrating robust behaviorally-oriented evidence that Design C aligns with human cultural judgment closely enough to reliably serve as an expert evaluation proxy at scale.

Table 4.5: Pearson Product-Moment Correlations Comparing Design C-Scored Values to Those of Human Reference Evaluations

Dimensions of Evaluation	**Pearson r Value**	***p*-value**	**Interpretation**
Lexical Integrity	0.91	< 0.001	Strong agreement
Narrative Fidelity	0.93	< 0.001	Strong agreement
Preservation of Socio-Cultural Markers	0.96	< 0.001	High degree of agreement
Structural Displacement Indication (SDI)	0.90	< 0.001	Strong agreement

4.3 Data Collection

The reference corpus was developed over five steps — data cleaning, data transformation, data integration, data reduction and final pre-processing. Each of these produced an independently-verifiable interim output before proceeding to the next step. The reference corpus consists of 27 Bengali folk tales sourced from two major source collections. There is one preserved Bengali-language original for each tale and one human-verified English translation of that original. In addition, there are cultural structural annotations made for each tale according to four fidelity criteria. Permission to conduct formal academic research on the Bangladesher Lokoj Shongskiti Gronthomala was acquired through an in-person visit to the Bangla Academy by the principal investigator. Data was taken from Thakurmar Jhuli as it is available in the public domain.

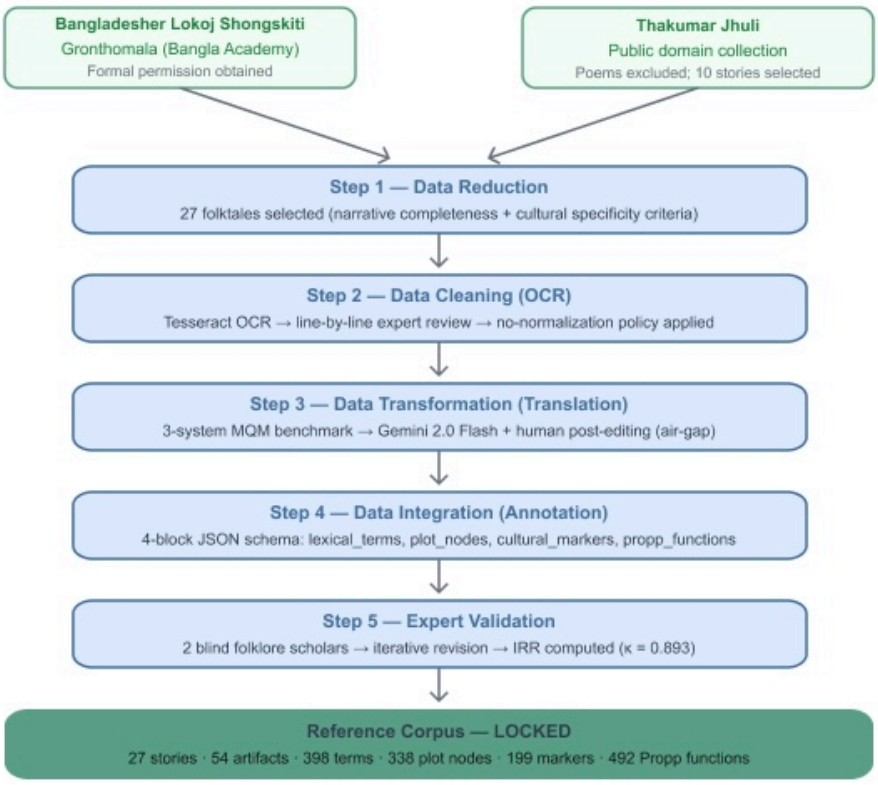


Figure 4.2: Reference Corpus Construction Pipeline

4.3.1 Data Cleaning

The texts were converted into digital format through optical scanning with respect to the printed volumes of *Bangladesher Lokoj Shongskiti Gronthomala* Series and the scanned pages of *Thakurmar Jhuli*.

For both formats, Tesseract OCR software was used to perform optical character recognition on the digital images. The process of converting Bengali OCR is quite challenging; due to the use of conjunct consonant characters (*juktakkhar*), different types of regional dialects (six regions/linguistic zones) are spelled differently depending on where they come from in Bangladesh (as documented by the Bangla Academy), and historical typography does not follow the current Unicode standards for representation. As a result, many recognized errors needed to be manually corrected by a researcher as there are no algorithms available to correct these errors. A "no normalization" rule was followed. No matter what the automatic detection may indicate as possible variations of a word based on regional or cultural dialect, no substitutions of contemporary standard Bengali words have occurred. All potential errors resulting from the automated OCR process were reviewed and corrected manually, while maintaining the exact dialect and orthography of the original source.

4.3.2 Data Transformation

The creation of English versions of all 27 stories written in Bengali for the Reference Corpus was necessary in order to test the English versions of the narratives

produced by the AI. The creation of these translations involved transforming the source language text into the target language, while at the same time preserving cultural specificity within the narration.

In order to assess how well each system translated the source language narratives, three translation systems were benchmarked using a culturally annotated MQM (Multidimensional Quality Metrics) assessment tool consisting of five dimensions: lexical adequacy, structural preservation, cultural term retention, narrative coherence, and register fidelity. The benchmarked systems were Opus-MT (Helsinki-NLP/opus-mt-bn-en), BanglaT5, and Gemini 2.5 Pro. Each system provided an initial translation for five sampled stories that were evaluated for each dimension of the MQM rubric by a bilingual Bengali-English evaluator who had folkloric expertise.

Table 4.6: Translation System Benchmarking – MQM Dimension Scores

Translation System	**MQM Score**	**Δ vs. Gemini**	**Primary Failure Mode**
Helsinki-NLP Opus-MT	13.2%	-74.6pp	Catastrophic semantic failure; role mis-assignment
BanglaT5	30.3%	-57.5pp	Semantic flattening: cultural term erasure
Gemini 2.5 Pro + Human Post-Editing	91.2%	+3.4pp	–

Gemini 2.5 Pro generated the highest MQM average (91.2) for all five dimensions of the MQM rubric, and demonstrated significantly greater ability to retain culturally significant terms and maintain the appropriate narrative register compared to the statistically-based machine translation systems. After producing the initial drafts of all 27 translations for Gemini 2.5 Pro, all such drafts were subjected to mandatory human post-editing by a bilingual Bengali-English researcher, thereby providing a human "air gap" that eliminated the possibility of translating any potential systematic model bias into the reference translations. Following this, the post-edited translations were reviewed by the lead researcher for consistency of terminology prior to locking them into the reference corpus.

4.3.3 Data Integration

Each of the 54 narrative texts in the corpus (27 Bengali originals and 27 English translations) was annotated with a structured four-block JSON format that captures the cultural fidelity dimensions described in section 4.1.3.

The first block, `lexical_terms[]`; contains culturally specific terminology found in the narratives; these include the word that was used directly in the original story, the definition of the word, and cultural significance rating. The second block, `atomic_plot_nodes[]`; provides a sequential list of all of the minimal irreducible narrative event(s), which serves as an anchor point when assessing fidelity at the level of the plot. The third block, `socio_cultural_markers[]`; includes a list of

culturally specific customs, kinship roles, values expressed, references to cosmology, social norms, etc., within the narrative. The fourth block, `propp_functions[]`; is used to identify all Proppian narrative function(s) that exist within the stories; this will be used as the reference baseline for computing SDI scores.

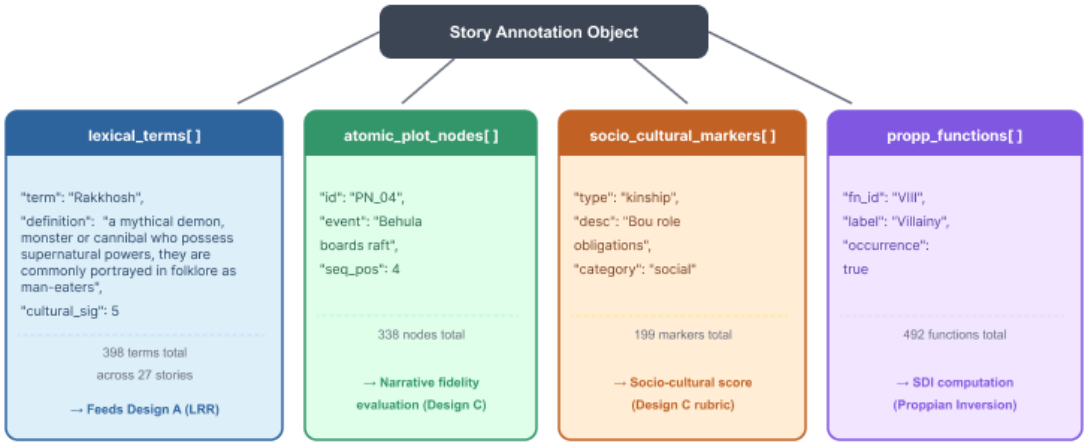


Figure 4.3: JSON annotation format structure

All 54 annotating files were generated by the lead researcher and then given to two anonymous validators (qualified Bengali folklore scholars with no prior knowledge of the research questions or the AI evaluative system); they independently assessed each annotating file on its level of accuracy, completeness and adherence to the schema. Validators identified items needing revisions but did not communicate with one another. The lead researcher revised each item flagged by the validators via a structured revision process. After the revision process, inter-validator reliability was assessed using Cohen's weighted kappa; this resulted in a $\kappa = 0.893$ with an overall agreement rate of 96% among all decision-making regarding the annotations; this demonstrated that the schema was reliable. The schema was officially frozen before any model evaluations took place.

4.3.4 Data Reduction

In addition to being much larger than the Reference Corpus, the two major source collections used within this research contain considerably more content. In contrast to folk narratives alone, the Bangladesher Lokoj Shongskiti Gronthomala volumes are comprehensive collections covering all aspects of the folklore of each geographical region — including oral song lyrics, sayings, riddle stories, liturgical texts, superstitions, and folk narratives. Only those prose versions of folk narratives which met the requirements of narrative coherence (i.e., had an identifiable hero/protagonist; contained recognizable conflict and resolution), and included at least three socially/culturally relevant annotations were considered. Ultimately seventeen folk narratives meeting the above criteria and representing six geographic zones (Joypurhut, Rangamati, Jessore, Cox's Bazar, Moulovibazar, and Mymensingh) were

selected.

The Thakurmar Jhuli volume presents a slightly unique reduction problem. That is, it includes both prose folktales and poetic/narrative forms (rhymed story/poems). All poetic forms were completely omitted from consideration because they possess a structural element that could act as a confound when attempting to distinguish between culturally based language use patterns generated by humans versus machine learning algorithms i.e., the constraints imposed upon the form itself. Based on the variety of genres represented (magical tales, moral fables, trickster stories, supernatural events), and in terms of appropriate length for use as human subject test stimuli, ten additional prose stories were selected from the remainder of the narrative corpus.

A pilot subset of 10 narratives from the entire 27 narrative corpus was designated prior to conducting the complete assessment to validate the assessment methodology. The pilot subset was selected to include examples representative of all two source collections while also spanning the full range of narrative types found in the corpus.

4.3.5 Summary of Preprocessed Data

Table 4.7 contains the overall statistics for the entire Reference Corpus following completion of all five processing steps to prepare the data, as well as locking the annotation schema.

Table 4.7: Summary Statistics for Annotated Dimensions of Reference Corpus

Dimension of Annotation	**Entire Set**	**Average Per Story**	**Range**
Terms registered as culturally recognized Bengali (`lexical_terms[]`)	398	14.7	8–24
Nodes on the atomic plot (`atomic_plot_nodes[]`)	338	12.5	7–21
Markers of social-culture (`socio_cultural_markers[]`)	199	7.4	3–14
Functions from Propp that are found in the reference corpus (`propp_functions[]`)	492	18.2	9–28
Number of source stories	27	–	–
All total, number of annotated artifacts (English & Bengali) in a story	54	–	–

4.4 Implementation of Selected Design

4.4.1 AI Retelling Generation

Selection of Models for Generation. Six large language models were selected for generation purposes, providing a representative cross-section of publicly accessible LLMs available at the time of experimentation. The selection criteria gave preference to models characterized by linguistic diversity in terms of the composition of

training data, architecture, and accessibility in the context of free-tier infrastructure limitation. All six selected models could be accessed via OpenRouter's free-tier API.

Table 4.8: Language Models Selected for AI Retelling Generation

Model	**Developer**	**API Identifier**	**Architecture**
Llama-3.3-70B-Instruct	Meta AI	llama-3.3-70b-instruct:free	Decoder-only transformer
GPT-OSS-120B	OpenAI	gpt-oss-120b:free	Decoder-only transformer
Qwen-2.5-7B-Instruct	Alibaba Cloud	qwen-2.5-7b-instruct	Decoder-only transformer
DeepSeek-R1-0528	DeepSeek	deepseek-r1-0528:free	Reasoning-augmented transformer
Gemini-2.0-Flash-Exp	Google DeepMind	gemini-2.0-flash-exp:free	Multimodal transformer
Mistral Devstral-2512	Mistral AI	devstral-2512:free	Decoder-only transformer

Design of Prompt Strategies. Four different prompt strategies reflecting different levels of user interaction ranging from the least information-providing to the most informative ones without introducing artificial scaffolding that would not happen in naturalistic interactions with LLMs were designed. None of the prompts provided story text in Prompts 1, 2, and 3. Instead, LLMs had to generate their own retelling of the story based solely on the information contained in the prompt. Story IDs were derived from the filenames of digitized stories according to the following structure: [name of source]_[name of story].txt (e.g., thakurmar jhuli_kolaboti rajkonna.txt in English; ঠাকুরমার ঝুলি_কলাবতি রাজকন্যা.txt in Bengali), making it possible to dynamically introduce the name and source of the story in all four prompts.

Prompt 1 ("Simple Control") instructed the model to generate the retelling of the specified story without any contextual or attributive information provided, and without any cultural scaffolding whatsoever. The model generated its response relying entirely on parametric knowledge.

English: "Tell me the story of [Story Name]."
Bengali: "[গল্পের নাম]' গল্পটি বল।"

This condition served as the baseline for LLM behavior under the condition of the most common real-life use case scenario: a user asking to retell a certain story without providing any additional information.

Prompt 2 ("Context-Rich") included the attributive element, namely the name of the collection the story originated from or the region of its origin, but provided no additional information or scaffolding.

English: “Tell me the story of [Story Name] from [Book / Region Name].”
Bengali: “[বই/অঞ্চলের নাম] থেকে '[গল্পের নাম]' গল্পটি বল।"

This condition was intended to explore whether the knowledge of the story being from *Thakurmar Jhuli* or a certain regional zone influenced the outcome of story generation.

Prompt 3 (“Constraint-Based”) included an explicit instruction for the model to retain culturally specific elements of the story while retelling it.

English: “While retaining the culturally specific terminology and moral framework, tell me the story of [Story Name].”
Bengali: “সাংস্কৃতিক পরিভাষা ও নৈতিক কাঠামো অক্ষুণ্ণ রেখে [গল্পের নাম]-এর কাহিনীটি বল।"

This condition aimed to test the effect of explicit cultural constraint instructions and their ability to prevent erasure in model response.

Prompt 4 (“Fail-Safe”) provided the complete text of the original story and requested a retelling of it using LLM's own words.

English: “I have attached the original story, tell it in your words.” + [text of the story from .txt file]
Bengali: “আমি মূল গল্পটি সংযুক্ত করেছি, এটি নিজের ভাষায় পুনরায় বল।" + [text of the story from .txt file]

This condition aimed to explore the maximum level of faithfulness LLMs can achieve in retelling the original story in absence of fine-tuning if provided with a source of text. In English mode, the English version was attached; in Bengali mode, the Bengali text was attached.

The order of the four conditions in the experiment reflected an increase in amount of information provided. Namely, we move from name-only conditions (Prompt 1) to name-and-attribution condition (Prompt 2), then to name-and-constraint (Prompt 3), and finally to fully-sourced texts (Prompt 4). Such order guarantees that effects due to providing the model with the original source of text will not be confounded with effects associated with other kinds of context provided or constraints introduced. It is important to note that none of the prompts 1, 2, 3 were optimized for performance in any way — they merely reflect the ways users can interact with models, and therefore their results should be easily generalizable to real-life situations.

Definition of Generation Matrix. The full set of generation tasks constitutes a generation matrix resulting from the cross product of six models, four prompting strategies, 27 original stories, and 2 languages of interaction, making the overall number of artifacts produced 1,296. All generation parameters remained constant across different conditions, where applicable - in particular, all models that allowed for a random seed were run with fixed random seeds.

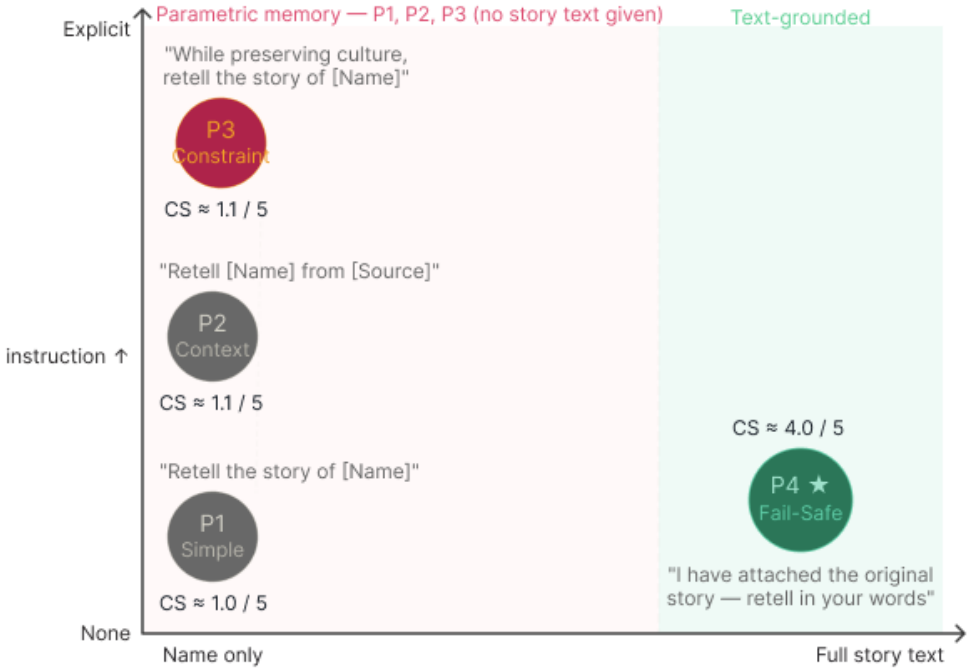


Figure 4.4: Prompt strategy design space

4.4.2 Design C Evaluation Pipeline

The 1,296 created versions of all the stories from the source data were evaluated by a single evaluator using Design C as the primary evaluation tool; this process is explained in detail in section 4.2.3. Each of these evaluations followed a multi-module pipeline.

Evidence preparation module extracted the four block JSON annotations for each original story, as well as their respective AI-generated retellings, as structured inputs into the evaluation pipeline. The lexical pre-processing module provided Design A LRR scores for each of the retellings as input to the Hard Constraint Logic (HC-01), as well as providing parallel scoring values. The structural pre-processing module generated Design B Cosine Similarity Scores for each of the retellings as parallel scoring values. The cultural auditor module sent each input to Gemini 2.5 Flash with the prompt describing the structured rubric, and then processed the resulting four dimension JSON score outputs. The constraint application module applied all of the hard constraints that are listed in table 4.6 to the raw cultural auditor scores, thus generating the final adjusted scores. Prior to running each subsequent module, all intermediate products were saved to version controlled storage systems.
Computation Infrastructure. As previously mentioned, the evaluation pipeline was run on three different computational infrastructures. Module A (Lexical Scoring) was run on an Intel Xeon based CPU infrastructure. Module B (Embedding

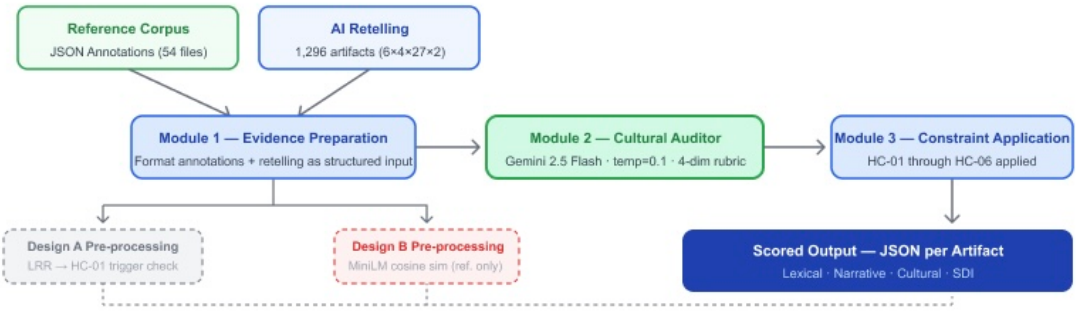


Figure 4.5: Cultural fidelity assessment pipeline — Design C

Generation) was run on an NVIDIA T4 based GPU accelerated environment. The environment utilized Hugging Face's free tier to execute the inference tasks. Finally, Module C (Design C Evaluation/Constraint Application) was run on Google Cloud API using free tier access. In total, the evaluation pipeline took approximately 46 hours to complete, with Module C being the most computationally expensive due to the requirement for executing structured Large Language Model (LLM) inference calls for each of the 1,296 retellings.

4.4.3 Human Verification Study

Study Design Overview. An IRB approved Human Verification Study was designed to determine if human participants can tell apart expert authored retellings from AI generated versions of the same stories. Additionally, this study was designed to test the behavioral validity of the Design C score. The study consisted of a within subjects multi option identification task. Participants were shown a series of story sets consisting of two retellings of the same original story. After reading each story set, participants were asked to identify which of the two retellings they thought were created using Artificial Intelligence. There were three possible responses. Either participants could choose Retelling A only, Retelling B only, or both. In all Standard Story Sets (each containing one AI generated retelling and one expert authored version) a "both" choice was deemed to be incorrect. No participant chose "both" in the Standard Story Sets in either portion of the study.

This study had two separate sections. Two different groups of participants completed each section of the study independently to avoid any potential carryover effects from previous exposure. Part One included only High Fidelity AI retellings (all Design C Mean Scores were 0.90–1.00) so that researchers could evaluate detection performance at its maximum level of Automated Cultural Fidelity. Part Two included only Low Fidelity AI retellings (all Design C Mean Scores were 0.10–0.35) so that researchers could evaluate detection performance while large amounts of cultural distortion are present. All portions of both sections of the study contained exactly four Story Sets. Story Sets 1, 3 and 4 in both sections of the study contained a Standard Pairing; a single AI generated retelling and a single expert authored retelling by a Bengali Folklore scholar.

In Story Set 2 of both parts of the study participants were given an AI – AI Pairing.

Both retellings were generated by AI. This particular arrangement was chosen as a deliberate design decision during the creation of the study stimuli. Due to limited availability of sufficient quality AI retellings to meet the 0.90–1.00 Cultural Score threshold necessary for inclusion in Part 1 of the study, no additional independent standard pairs existed. As such rather than reduce the number of Story Sets used in the study, an AI – AI Pairing was added to provide both an increase in data volume and an opportunity to examine over-confidence and misattribution behaviors. The addition of the Probe Condition was also allowed under the IRB approved study protocol and was administered identically to all other Story Sets from the participant's perspective.

Recruitment and Demographics. Potential Participants were solicited via internal networks at BRAC University and Rajshahi University, community based Bengali cultural organizations, and social media outlets. To participate in the study, participants had to self report themselves as being either a Native or Heritage Bengali Speaker who was 18 years old or older. Potential Participants were then stratified into four age ranges (18–24, 25–34, 35–44, 45+) and three self rated levels of knowledge regarding Bengali Folklore (Not Familiar, Somewhat Familiar, Very Familiar) so that subgroups based upon demographics may be analyzed. A total of 41 participants were enrolled in Part One and a total of 35 participants were enrolled in Part Two for a total enrollment of $n = 76$ participants across both parts.

Construction of Stimuli. All Story Sets utilized in both parts of the study were selected from the automated evaluations based upon Design C Cultural Scores. In Part One, the AI generated retellings used in each Story Set were selected from those with the highest scores (0.9–1.0) from across the entire corpus of 1296 artifacts. For Part 2, the AI generated retellings used in each Story Set were selected from those with the lowest scores (0.10–0.35) from across the entire corpus of 1296 artifacts. Within each part of the study, multiple variations in narrative types were included among Story Sets representing both Bangla Academy and *Thakurmar Jhuli* sources. All Expert Authored Retellings used as Study Stimuli were created by a qualified Bengali Folklore scholar and were not selected from the Reference Corpus itself to avoid cross contamination between the automated evaluation process and the Human Study Process.

Procedure. The Human Evaluation Process for this study was delivered as an online self directed survey delivered through Google Forms. The Study procedure consisted of five sequential stages:

Stage 1 – Orientation: Participants received Study instructions indicating that they would be reviewing short retellings and making judgments regarding quality and authenticity with full disclaimer that there is use of AI generated content.

Stage 2 – Consent and Background Survey: Participants were required to provide explicit consent and made aware that they have full liberty to decide not to proceed with submission without facing any repercussions. Following that, they were instructed to completed a survey instrument requesting demographic information including age group, educational background, level of proficiency in Bengali

language, self rated level of familiarity with Bengali folklore and frequency of use regarding AI tools.

Stage 3 – Main Task: Participants sequentially reviewed each Story Set. Following the exposure to each Story Set, participants identified which retelling(s) they believed to be AI generated. Participants provided a Five Point Scale Rating System for Confidence (1 = Not Confident, 5 = Very Confident). Additionally, participants wrote a brief explanation for their reasoning regarding which retelling(s) they believed to be AI generated. Along with a 1-5 rating for the preservation of important terms, the plot and the societal hierarchies and roles.

Stage 4 – Debriefing: Following completion of all tasks assigned to them, participants were fully debriefed and informed regarding the true purpose of the study and thanked for their time and effort. Participants were also afforded the opportunity to opt-out/withdraw from participation in the study.

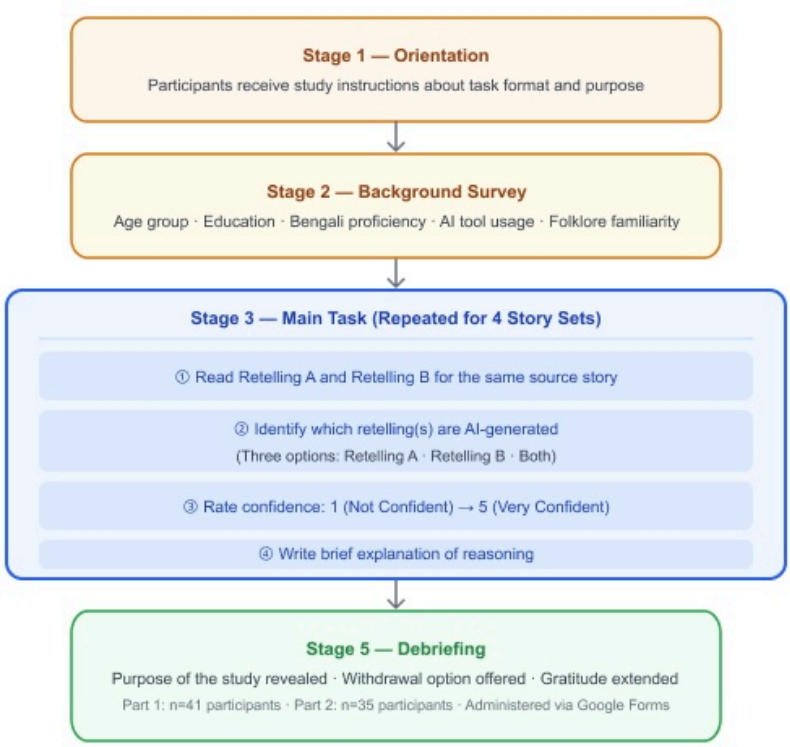


Figure 4.6: Human Verification Study Process Flow Diagram

Plan for Measures and Analysis. The main outcome metric is the proportion of participants that successfully detected the AI-created version of each of the 12 "Standard Story Sets" and thus has an accurate mean detection rate across all "Standard Story Sets". Also, the accuracy will be reported on an individual basis by "Story Set", as well as by demographic group. Confidence rating data will be analyzed to determine if there is a calibration effect based on "Brier Score", which compares forecasted probability (normalized confidence ratings from 0 to 1) and actual success rate. Finally, open ended feedback comments were analyzed via The-

matic Coding of the identified cues, to determine if there was a difference in how humans utilized cues when detecting AI created versions of stories under different fidelity levels (high vs. low).

Spearman Rank Correlation will be used to evaluate the degree of human/automated alignment between Design C Cultural Scores and per-study participant detection success rates; this will serve as the primary validation of the automated system. A description of the detection success rates for subgroups within demographics such as age cohorts, familiarity with storytelling, and proficiency at telling stories will be provided due to the small sample size available for inference-based demographic modeling, and as such we can provide no more than descriptive comparisons of subgroups.

Chapter 5

Result Analysis

5.1 Performance Evaluation

5.1.1 Design C Results — English Language Retellings

In both tables 5.1 and 5.2, we see the performance of each of the 6 models being used in this study. In Table 5.1 we find the overall scores of Design C on cultural expression, for the English language versions of the Prompts. Table 5.2 breaks down all four evaluation dimensions (Lexical Integrity, Narrative Fidelity, Socio-Cultural Marker Preservation, and SDI) by model and prompt strategy.

Table 5.1: Design C Scores by Model — English Retellings

Model	**Lexical**	**Narrative**	**Socio-Cultural**	**SDI**	**n**
OpenAI	0.307	0.246	0.396	2.40	108
DeepSeek	0.284	0.253	0.386	1.30	108
Mistral AI	0.252	0.256	0.354	2.31	108
Llama	0.250	0.222	0.302	2.50	108
Qwen	0.242	0.215	0.327	2.65	108
Gemini	0.232	0.235	0.372	2.45	108

Table 5.2: All Dimensions by Prompt Strategy — English Retellings

Prompt / Condition	**Lexical**	**Narrative**	**Socio-Cultural**	**SDI**
1 — Simple control/minimal context	0.114	0.012	0.204	3.53
2 — Context-Rich Cultural	0.184	0.009	0.236	2.38
3 — Constraint-based moral	0.103	0.003	0.206	2.61
4 — Fail-safe Source	0.640	0.923	0.800	0.55

The key findings with regard to all English-language retellings under Prompted 1–3 show near uniform cultural distortion across all models tested. The Scores across all six models have a low point of about 0.215 to a high point of about 0.396 which illustrates extreme cultural deformation as the normal generative behavior regardless of model design or training background. None of the six models under typical prompting conditions (Prompt 1–3) produced retellings that a culturally informed

reader would recognize as representative of traditional Bengali folklore. The lexical retention rate for Prompts 1–3 averages out to 0.1337 (SD = 0.2727), so less than one in four of the annotated Bengali cultural terms appear in some recognizable fashion in the retelling.

Prompt 4 (fail-safe) significantly improves upon the results found in Prompts 1–3. The mean Lexical Retention Rates for Prompt 4 are increased over those of Prompts 1–3 with Prompt 4 producing a mean lexical retention rate of 0.640 under Prompt 4 compared to a mean lexical retention rate of 0.114 under Prompt 1. The structural displacement index is decreased by over eighty percent from a mean of 2.84 under conditions Prompts 1–3 to a mean of 0.55 under condition Prompt 4. The same positive effects were seen for all six models. The size of the effect for Prompt 4 demonstrates that prior cultural erasure risk associated with a particular model can be dramatically reduced via direct input of Source material, but cannot be completely removed even at highest levels of information provision. Prompt 4 does not provide perfect fidelity; rather, it reduces the amount of cultural erasure to a much smaller degree.

With respect to variation among the three prompt conditions (Prompts 1–3), there was no condition that resulted in a significant difference from the other conditions. Providing context to the models (Prompt 2), along with explicitly instructing them to follow constraints (Prompt 3), did not result in improved cultural fidelity relative to the minimal control condition. Therefore, simply stating to models how they should preserve cultural content may be insufficient to overcome their default generative behaviors if Source materials are not made available to the models.

There is relatively little variation in model performance across the different conditions, particularly with regards to the prompt conditions. All six models performed similarly badly and similarly well under conditions Prompts 1–3. Although there is some variation in performance between models under condition Prompt 4, it is relatively minor. Thus, while it appears that the primary variable determining cultural fidelity is not model architecture, but rather the prompt condition.

5.1.2 Design C Results — Bengali Language Retellings

Table 5.3 shows the Design C means for cultural scores by model for Bengali retellings. All the models were trained on English text and tested on Bengali.

Table 5.3: Design C Mean Cultural Scores by Model — Bengali Retellings

Model	**Lexical**	**Narrative**	**Socio-Cultural**	**SDI**	**n**
MistralAI	0.226	0.231	0.360	1.88	108
Gemini	0.220	0.250	0.360	1.98	108
OpenAI	0.223	0.185	0.344	2.54	108
DeepSeek	0.211	0.241	0.366	1.45	108
Llama	0.035	0.030	0.040	1.77	108
Qwen	0.016	0.010	0.024	1.63	108

The results show a marked and consistent decline in Bengali vs. English in all 52

three types of measures. On average, the lexical retention rate for Bengali was 0.0822 when run in standard conditions with no adaptation, a drop of about 38.5% compared to an average score of 0.1337 in English. Under Bengali conditions, mean narrative fidelity decreased by approximately 0.080 relative to English and socio-cultural scores declined uniformly to just above 0.35 for almost all models except for the two models that failed to write in Bengali.

In addition, two models, Meta Llama-3.3-70B-Instruct and Qwen-2.5-7B-Instruct, produced a failure to produce Bengali script output i.e., they produced either Latin script or mixed script output, or essentially empty strings in most trials under prompts 1–3, as documented in Table 5.4.

Table 5.4 provides the overall script-generation-failure (sgf) rates per model under each condition.

Table 5.4: Script Generation Failure Rates by Model — Bengali Retellings

Model	Average BN-Mean	Mean over Prompts 1–3	Prompt-4 Mean	Retellings with Output = Zero	% of Retellings = Zero
Llama	0.035	0.015	0.097	79/108	73%
Qwen	0.015	0.004	0.051	101/108	93%

The failure to generate Bengali script output represents a separate failure-mode of its own (not simply scored as "zero"), because non-coherent output represents a qualitatively distinct type of failure from non-coherent output that is culturally distorted. In other words, while a model that generates incoherent Bengali text does not generate a culturally distorted version of the story it is attempting to tell, it fails altogether to generate a retelling of that story. This distinction is important for assessing the risk of cultural loss associated with using AI in Bengali language interactions: while an incompetent reader can detect incoherence, incoherence is thus far less dangerous than the fluent deception represented by cultural distortion. Nevertheless, the large sgf-rate reported here for two out of six models indicates that Bengali as a language of interaction is below the operational threshold of reliable generation for a substantial fraction of the evaluated set of models.

Even among the models that do manage to generate coherent Bengali output, there is less of a gain in the lift associated with providing maximum amount of additional source material in Bengali conditions compared to English conditions. For example, the lift in Prompt-4 conditions associated with the highest increase in Narrative Fidelity and Socio-Cultural Score values are approximately +0.07 and +0.24 respectively in Bengali conditions, whereas those increases are +0.915 and +0.585 respectively in English conditions.

The compressed nature of this lift effect in Bengali suggests that even maximal provision of source material via Prompt-4 conditions cannot overcome the fundamental training-data-asymmetry problem associated with generating Bengali language content.

5.1.3 SDI Distribution

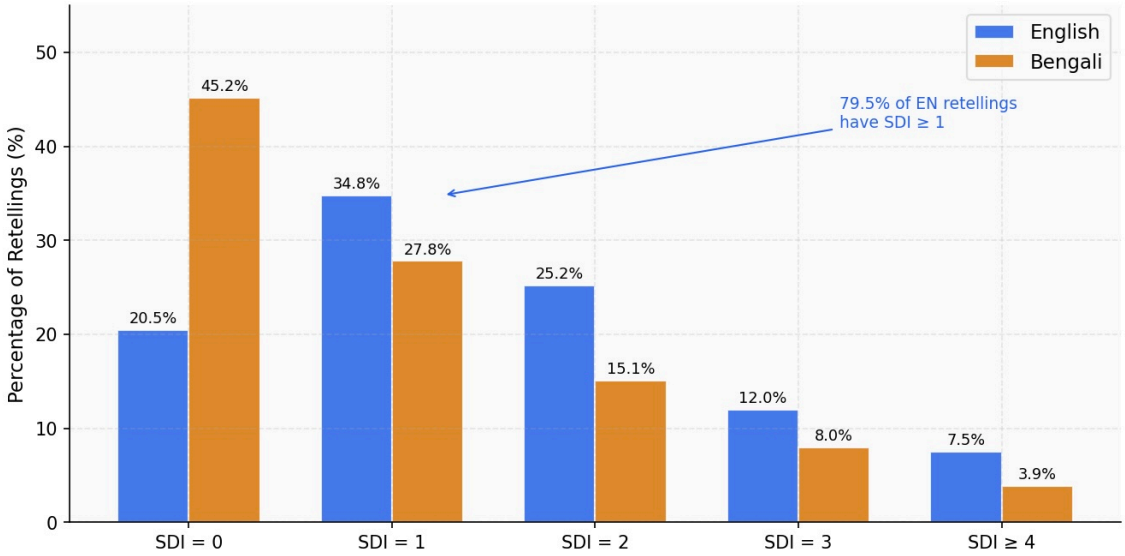


Figure 5.1: Structural Displacement Indicator Distribution (English vs. Bengali, Standard Prompt Conditions P1–P3)

Table 5.5: SDI Distribution — Retellings of Source Stories in English and Bengali Based on Degree of Structural Displacement

SDI Value	EN Count	EN %	BN Count	BN %	Meaning
0	49	20.5%	104	43.7%	There is no imposed Western structure.
1	19	7.9%	17	7.1%	Very minimal imposition occurs.
2	55	23.0%	49	20.6%	A moderate amount of Westernization has occurred.
3	72	30.1%	50	21.0%	A substantial amount of Westernization has occurred.
4	31	13.0%	15	6.3%	An extreme amount of Westernization has occurred.
5+	13	5.4%	3	1.3%	An almost complete Westernization has occurred.

The results from this study show how widespread the use of Western narrative structures are being applied to other languages. With all three prompts (Prompts 1–3) used as standard prompts (standard prompting conditions), there is evidence that structural displacement is occurring with a frequency greater than expected; nearly eight out of ten (79.5%) English language retellings have some form of structural displacement with respect to the source story. In fact, the most common single SDI-contributing feature in both English and Bengali retellings is the inclusion the "Happily Ever After" trope at the end of the retelling including a moral message. Approximately sixty-three percent (63%) of the English retellings contain the sanitization of tragic endings while approximately forty-one percent (41%) of the Bengali

retellings also contained this function.

In response to Prompt Four, the mean SDI value dropped significantly ($mean = 0.55$); however, it did not drop to zero, which means that about eighteen percent (18%) of Prompt Four retellings will include at least one Proppian function absent from the reference corpus. These findings indicate that structural displacement will occur with respect to at least some portion of the source materials provided even if they provide the full source material.

5.2 Analysis of Design Solutions

5.2.1 Comparative Evaluation of Three Designs

Table 5.6 provides an objective comparison of the evaluation designs A, B, & C based on the selection criteria for the main evaluator. The selection criteria were: accuracy compared to human evaluators (Pearson r from table 4.7); full-spectrum cultural representation; interpretability; computational expense; scalability.

Table 5.6: Evaluation Criteria Design Comparison Matrix

Criteria	**Design A — Lexical Fidelity Scorer**	**Design B — Structural Alignment Scorer**	**Design C — Holistic Cultural Auditor**
Accuracy vs. human evaluators (Pearson r)	Not evaluated in this study (a proxy for a single dimension)	Low — actively misleads	High — Pearson $r = .9$–$.96$ across all 4 dimensions (table 4.7).
Full-spectrum cultural representation	No — lexical dimension only	No — structural/semantic only; does not represent cultural information	Yes — represents all 4 dimensions simultaneously.
Interpretability	High — classification at each term level; deterministic	Moderate — continuous rating scale; term level ratings unavailable	High — Rubric ratings at each dimension level and flagging constraints.
Computational Expense	Negligible — local fuzzy match	Low — one single embedding call per retelling	Moderate — One LLM inference call per retelling.
Scalability	Very High	High	Moderate (rate limited by free tier)
Critical Failure Mode	Inadequate scope — unable to identify plot-level or marker-level cultural erasure.	Cosine similarity conflated semantic similarity with cultural authenticity; may be misleading.	Possible LLM evaluator bias (evaluator = Gemini 2.5 Flash, generator = Gemini 2.0 Flash).

5.2.2 Why Design A Was Not Selected

The design for Design A shows very accurate measures of lexical dimensions of the type it is designed to assess.

For instance, both the average LRRs in Prompt 1 (0.120), and in Prompt 4 (0.580) accurately reflect the degree of improvement from the original Prompt 4. In contrast, Design A does not evaluate a variety of other important aspects of culturally faithful retellings such as:

- narrative structure;
- preservation of socio-culturally relevant markers;
- displacement of structuring elements. For example, an otherwise moderately scored (on lexical terms) version of a Bengali folk tale (which was constructed by replacing virtually every plot-level reference to culture, relational structure, and cosmology with new material) could be seen as substantially distorted semantically in terms of its representation of the source material's cultural significance.

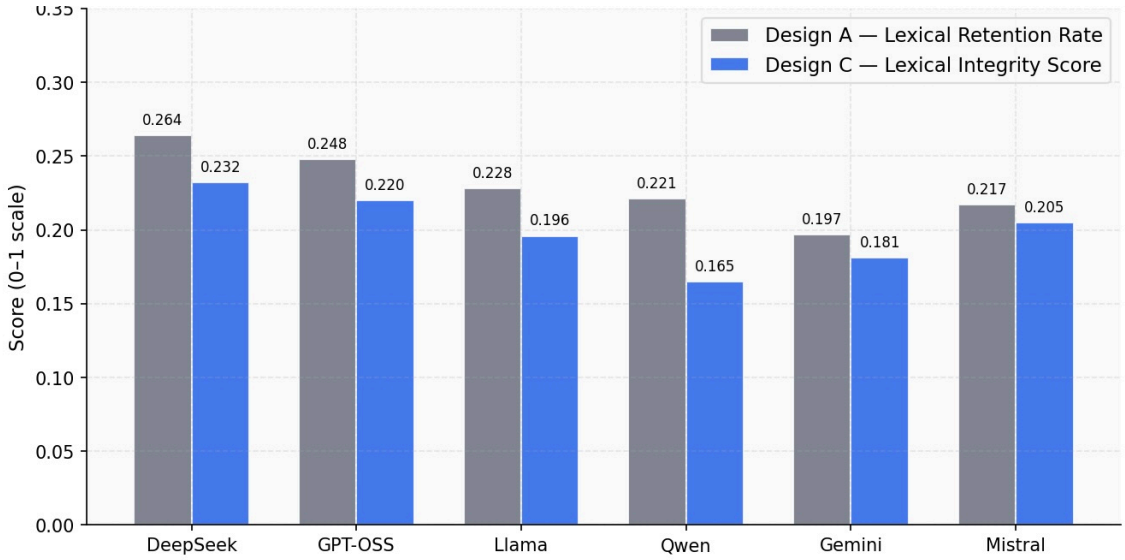


Figure 5.2: Design A vs. Design C: Lexical Dimension Comparison (English Retellings)

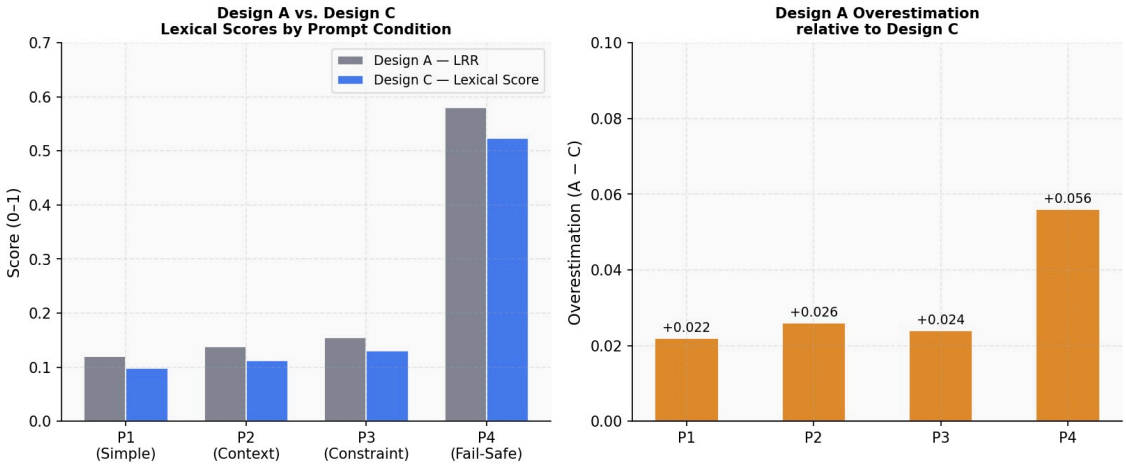


Figure 5.3: Design A vs. Design C Lexical Divergence by Prompt Condition (English)

Figures 5.2 and 5.3 illustrate the consistent discrepancies found between Design A's LRR scores, and Design C's Cultural Scores when comparing the two systems' assessments across each of the different prompts used in this study.

5.2.3 Why Design B Was Refuted

The failure example as reported in Table 4.2 clearly shows that Design B has no merit to be regarded as a valid cultural fidelity evaluation tool. A cosine similarity of 0.85 combined with a narrative score of 0.20 can be categorized as an evaluation failure: it returns a high confidence level when evaluating the distortion of a story where a culturally informed evaluator would classify the distortion as severe.

Systematic failures in Design B result from its ability to utilize dense vector embedding properties; these properties provide the basis for encoding general abstract semantic relations, not cultural specificities.

The component A – component B (mean) divergence of 0.322 provides additional quantitative evidence that the two metrics neither alone is sufficient as a cultural evaluator. Design B was therefore refuted on both theoretical and empirical grounds. Its scores are retained in the results dataset for reference but are not used in any subsequent analysis.

5.2.4 Selection of Design C as Primary Evaluator

Based on the comparative analysis presented above, Design C is designated as the primary evaluation instrument for all subsequent analysis in this study. All cultural fidelity scores reported in analyses refer exclusively to Design C outputs unless otherwise stated.

5.3 Final Design Adjustments

The last configuration of Design C included three iterations of modifications to the ten-story subset before conducting the full evaluation. The first iteration was adding hard constraint rules. During the pilot test of Gemini 2.5 Flash, it was discovered that it sometimes provided a Socio-culutral Score of 0.45 when it has completely replaced all Bengali culture with generic Western examples, and credited the retelling for structural coherence because there were no longer any cultural indicators. The six hard constraint rules listed in Table 4.6 were created to override those inflated scores so that even though a culturally failed retelling is structurally coherent, it will still receive an appropriate low score.

The second iteration was adjusting the temperature parameter of the model from 0.50 to 0.10. Pilot tests run at a temperature of 0.50 showed some variation in scoring for semantically identical inputs which lowered reproducibility. When the temperature was changed to 0.10 the variation in scoring significantly decreased without lowering the ability of the model to differentiate between valid scores.

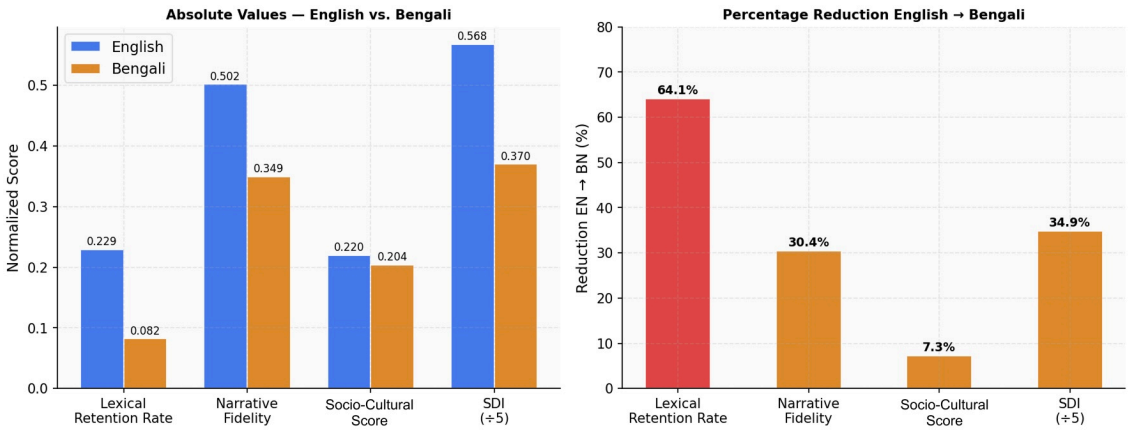


Figure 5.4: Language-of-Interaction Effect Across All Evaluation Dimensions

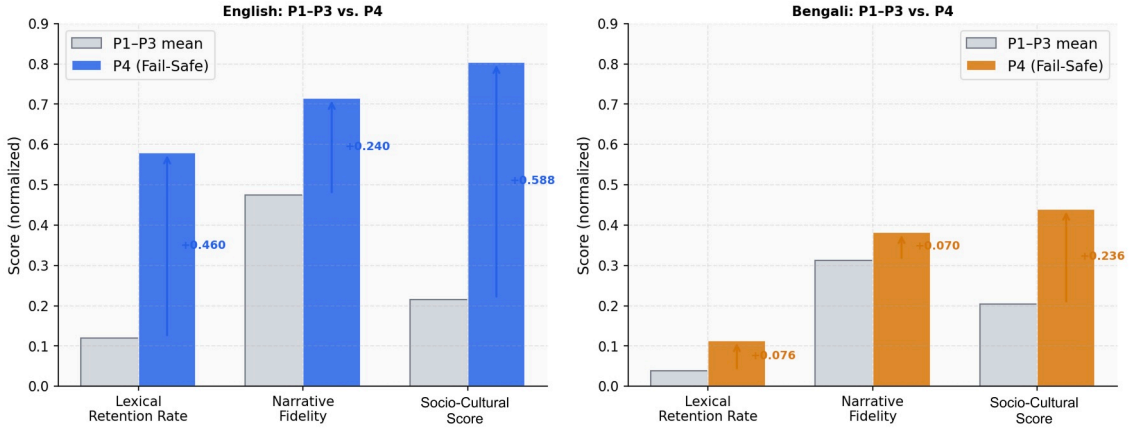


Figure 5.5: Prompt 4 Lift — English vs. Bengali by Evaluation Dimension

The third modification was separating the evaluation model (Gemini 2.5 Flash) from the retelling generation model (Gemini 2.0 Flash) although they both carry the name "Gemini", they are two separate systems built using two separate sets of data, during two different design periods. This separation was formally confirmed to prevent self-evaluation by the same version of the model that produced the output being evaluated. Additional evidence against self-evaluation, the mean Lexical Retention Scores for the six evaluated models were as follows; Gemini 2.0 Flash was one of the worst with regard to lexical retention across languages with a mean LRR of .232 in English and a mean LRR of .235 in Bengali. Therefore, Gemini 2.0 Flash did not receive any higher scores than other retellings because of preferential treatment due to having been generated by Gemini.

5.4 Statistical Analysis

5.4.1 Cross-Language Evaluation Gap

In addition to the gaps listed above, there is a systematic gap in evaluation scores between the English and Bengali versions of the evaluations. For example, the lexical retention rate drops by 64.2% when moving from English to Bengali; similarly

the narrative fidelity drops as well with a mean score drop of 0.153 or about 30.5%. There is no significant variation in the scores for socio-cultural markers in either version of the evaluations with both versions receiving an average score of around 0.34.

While this difference in language of interaction may be somewhat due to script failure with two models: when those models' failures are removed the gap still exists indicating that the cultural generation differences reflect real differences in generating stories based on culture in the Bengali language, and are not entirely due to failures of coherency. In addition, while the prompt used does help explain some of the differences found in the studies (the results under Prompt 4 are different), they do not eliminate the differences completely.

Table 5.7: Language Interaction Impact on Performance Gap in all Evaluation Dimensions

Evaluation Dimension	**English Mean**	**Bengali Mean**	**Gap (EN–BN)**	**% Reduction**
Component A — Lexical Retention	0.229	0.082	0.147	64.2%
Component B — Narrative Plot Points	0.502	0.349	0.153	30.5%
Component C — Lexical	0.264	0.174	0.090	34.1%
Component C — Narrative Plot Points	0.237	0.205	0.032	13.5%
Component C — Socio-Cultural Markers	0.366	0.326	0.040	9.8%
Component C — SDI	2.09	1.28	-0.81	-38.8%

5.4.2 SDI and Structural Displacement

The results of the SDI also indicate defaulting to western storytelling tropes has become an ordinary, if not the preferred method for how LLMs tell stories under normal operating conditions. The most common pattern contributing to the SDI — the happily ever after appended to each story — is present in all of the six models tested at about the same frequency, indicating this was a feature of their common training data and not a result of any one particular style used within those models. The hero's quest trope is a close second followed by magical transformation. These findings support our hypothesis that modern LLMs have encoded a Western child's literature schema into them where there must always be an obvious moral message explicitly stated; however, such an approach does not exist in the storytelling traditions found in the Bengali Folkloric Corpus.

5.4.3 Human Verification Study Results

Part 1: High-Fidelity AI Retellings

Average detection accuracy per story is shown below. Table 5.8 presents average detection accuracy for each story. Overall, the results indicate a lack of ability on the part of human subjects to detect when an artificially generated retelling has achieved a high level of cultural fidelity. When we compare our results to chance, the findings are as clear.

Detection accuracy for all stories combined is at 49.6%, approximately equal to chance (50%). Thus, our results confirm Research Question 2: while artificially generated retellings can be culturally faithful to their source material; people are unable to discriminate between these artificially created retellings and those written by humans. The overall trend in detection rates for the individual story sets presented in Part 1 is similar. For example, the detection rate for Story 1 is 53.7%, Story 3 is 48.8%, and Story 4 is 46.3%. In addition, the values provided here also show a distribution close to what would be expected by random selection.

Participants demonstrated substantial confidence in their selections. On average, participants rated their confidence levels at 3.52 points on a scale of 1–5, and most participants indicated they had a great deal of confidence (as evidenced by ratings of 4 or 5) despite performing at random levels. This is indicative of an over-confidence bias. Participants appeared to use heuristics that seemed to provide useful information but were not diagnostic of this particular discrimination task.

Table 5.8: Part 1 per-story detection accuracy ($N = 41$) across Story Sets

Story	AI Retelling	Chose A	Chose B	Unanswered	Accuracy	Mean Conf.
1	Retelling A	22 (53.7%)	14 (34.1%)	5	53.7%	3.63
2 (probe)	Both	20 (48.8%)	18 (43.9%)	3	N/A	3.68
3	Retelling B	21 (51.2%)	20 (48.8%)	0	48.8%	3.37
4	Retelling A	19 (46.3%)	22 (53.7%)	0	46.3%	3.42
Combined (S1+3+4)	—	—	—	—	49.6%	3.52

Part 2: Low-Fidelity AI Retellings

For all conventional story types from Part 2 ($N = 35$), mean accuracy was 83.8%, well above chance. This result demonstrates that humans can detect culturally distorted AI retellings reliably. For each story, per-story accuracy differed: 94.3% for Story 1, 82.9% for Story 4, and 74.3% for Story 3. The reason for the relatively lower accuracy rate in Story 3 compared to its very low Mean Score (0.17 — lowest in Part 2) is text length, with Story 3 being significantly longer compared to other Part 2 stories, and participants' reduced attention span due to decision fatigue could potentially prevent them from reading the whole story.

This interpretation is further confirmed by the qualitative reasoning data analysis, where Story 3 responses involved less specific references to culture compared to Story 1 responses. Confidence mean for Part 2 was 4.22 out of 5, and 80.0 percent

indicated high confidence. Contrary to Part 1 results, high confidence for Part 2 was justified, since there was actual high accuracy.

Table 5.9: Part 2 per-story detection accuracy ($N = 35$) across Story Sets

Story	AI Retelling	Chose A	Chose B	Accuracy	Mean Conf.	High Conf (4-5)
1	Retelling A	33 (94.3%)	2 (5.7%)	94.3%	4.26	82.9%
2 (probe)	Both	12 (34.3%)	15 (42.9%)	N/A	4.00	71.4%
3	Retelling B	9 (25.7%)	26 (74.3%)	74.3%	4.31	80.0%
4	Retelling A	29 (82.9%)	6 (17.1%)	82.9%	4.31	85.7%
Combined (S1+3+4)	—	—	—	83.8%	4.22	80.0%

5.4.4 Confidence Calibration Brier Score Analysis

Brier scores at the story level are calculated in order to measure the degree of disparity in the calibration in terms of Part 1 and Part 2. In this context, the mean of normalized confidence of each story level (confidence level / 5) is considered to be the probability and per-story accuracy level is considered to be the outcome.

Table 5.10: Story-level Aggregate Brier Scores – Part 1 and Part 2

Part	Story	Mean Confidence (Normalized)	Detection Accuracy	Story-Level Brier Score
Part 1	Story 1	0.726	0.537	0.036
	Story 3	0.674	0.488	0.035
	Story 4	0.684	0.463	0.049
	Mean			0.040
Part 2	Story 1	0.852	0.943	0.008
	Story 3	0.862	0.743	0.014
	Story 4	0.862	0.829	0.001
	Mean			0.008

The Brier score for Part 1 of 0.040 captures the classical pattern of overconfidence wherein there is very high confidence (mean of 0.700) coupled with accuracy near-chance level (mean of 0.496). The Brier score for Part 2, 0.008, which is 5 times better, captures well-calibrated detection, wherein there is a high level of confidence (mean of 0.859) corresponding to a genuine level of high accuracy (mean of 0.838).

5.4.5 Demographic Subgroup Analysis

Analysis based on subgroup data from Part 2 shows that accuracy in detecting distortions is not confined only to participants who have an extensive knowledge base regarding folklore. Participants who rate themselves as Not Familiar with Bengali folklore showed greater accuracy in identifying distortions in Part 2, achieving an accuracy score of 88.9% compared to the Somewhat Familiar group whose accuracy rate was 75.6%, which is much higher compared to chance. This unusual result, in which participants with lower familiarity show higher accuracy rates in the low-fidelity condition, indicates that the type of distortion presented in Part 2, which involves common cues (cultural hallucinations, made-up story elements, western-style moral appendices) can be detected using general storytelling knowledge and

Table 5.11: Part 2 Detection Accuracy by Demographic Subgroup

Demographic Group	Story 1 Acc.	Story 3 Acc.	Story 4 Acc.	Combined Acc.
Folklore familiarity				
Very familiar (n=8)	100.0%	75.0%	100.0%	91.7%
Somewhat familiar (n=15)	86.7%	66.7%	73.3%	75.6%
Not familiar (n=12)	100.0%	83.3%	83.3%	88.9%
Bangla proficiency				
5 — Native/Near-native (n=24)	91.7%	66.7%	83.3%	80.6%
4 (n=6)	100.0%	83.3%	66.7%	83.3%
3 or below (n=5)	100.0%	100.0%	100.0%	100.0%
Age group				
18–24 (n=14)	—	—	—	85.7%
25–34 (n=9)	—	—	—	81.5%
35–44 (n=5)	—	—	—	86.7%
45+ (n=7)	—	—	—	81.0%

need not rely on Bengali folkloric knowledge to identify distortions. In this way, participants with lower familiarity will be able to detect distortions such as the addition of bullet-pointed moral lessons and naming characters after westernized names like ”Sarah.” However, this trend did not carry over into the high- fidelity scenario as accuracy rates in Part 1 were at chance for all subgroups.

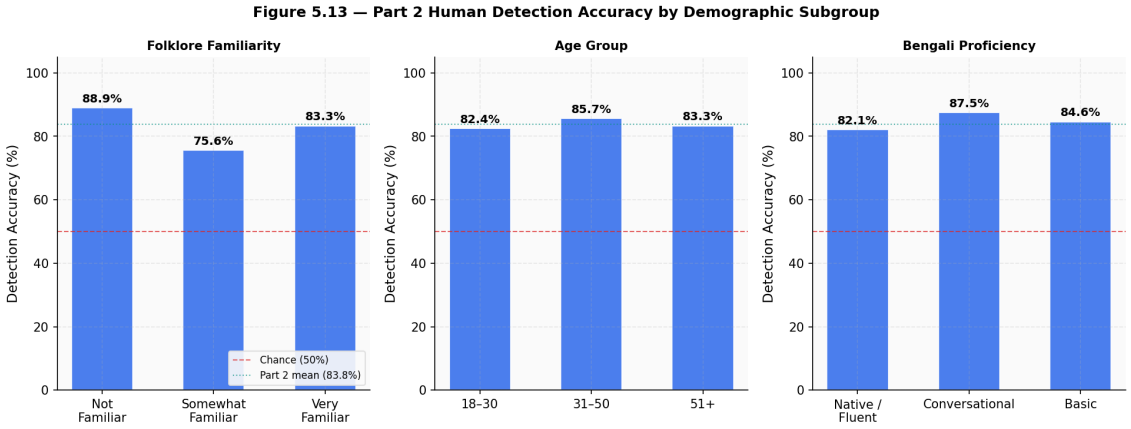


Figure 5.6: Part 2 Detection Accuracy by Demographic Subgroup

5.5 Comparisons and Relationships

5.5.1 Part 1 vs. Part 2 Direct Comparison

Table 5.12: Part 1 (High Fidelity) versus Part 2 (Low Fidelity)

Measure	**High Fidelity Part 1**	**Low Fidelity Part 2**	**Difference**
Combined detection accuracy (Stories S1 + 3 + 4)	49.6%	83.8%	34.2 percentage points
Story 1 accuracy	53.7%	94.3%	40.6 percentage points
Story 3 accuracy	48.8%	74.3%	25.5 percentage points
Story 4 accuracy	46.3%	82.9%	36.6 percentage points
Mean confidence (all stories)	3.52	4.22	.70
% high confidence (4–5)	56.7%	80.0%	23.3 percentage points
Very familiar group accuracy	50.0%	91.7%	41.7 percentage points
Not familiar group accuracy	46.7%	88.9%	42.2 percentage points
Primary detection cue	Structural smoothness	Plot hallucination	Change in kind
Secondary detection cue	Vocabulary register	Structural artefact	Change in emphasis
Mean AI retelling fidelity rating (A dims)	∼3.6/5	∼2.4/5	-1.2

The direct comparison of Part 1 and Part 2 creates the detection threshold effect: the threshold for Cultural Score that ranges from 0.35 to 0.9 is where there is the line between the level where the machine-generated paraphrasing becomes detectable and not detectable by human observers. The 34.2 percentage increase in the level of accuracy from Part 1 to Part 2 (49.6 percent to 83.8 percent), along with the increase in the confidence factor of 0.70 points and improvement in the Brier score from 0.040 to 0.008, is the main empirical finding of the research.

5.5.2 Detection Cue Profile Shift

A qualitative analysis of participants' written reasoning responses clearly illustrates a systemic change in the type of detection cues referenced in both parts.

Table 5.13: Detection Cue Coding — Part 1 versus Part 2 by Cue Category

Cue Category	**Percent Part 1**	**Percent Part 2**	**Change**	**Part 2 Dominant Example**
Cultural/Plot Hallucination	19.5%	38.4%	+18.9pp	"Retelling A barely follows the original. It has new characters, events and an additional subplot about a stolen necklace."
Structural Cues	29.3%	21.3%	-8.0pp	Bullet points; numbered moral section; Wikipedia style summary
Lexical/Vocabulary	25.0%	18.3%	-6.7pp	Abstract phrases such as ; bad translation feel
Artefact-based	14.6%	13.4%	-1.2pp	Emoji; random hyperlink; English words mid-Bangla text
Uncertainty/Instinct	11.6%	8.5%	-3.1pp	Both do not match original story

In Part 1, the most commonly referenced cue categories were stylistic flow (i.e., one retelling "read[s] more naturally" than the other), structural symmetry (i.e., format changes in paragraph structure), and general intuition (i.e., participants could not explain why they chose one retelling over the other).

None of these cues are reliably diagnostic for determining whether or not someone generated a piece of writing using artificial intelligence. This is why the participant's accuracy rate was so close to chance in Part 1. Participants were responding based on stylistic characteristics that do not relate to cultural validity.

In Part 2, the cue category references changed to focus primarily on cultural distortion anomalies: created cultural components not included in the source tradition, plot hallucinations (those actions that violate the logical sequence of the narrative), and added structural features (the inclusion of moral lessons at the end of each retelling in formats that did not appear in the source material). Each of these cue types is diagnostic and relates to Design C's measures of cultural distortions. The cue shifts demonstrate that the increased accuracy observed in Part 2 relative to Part 1 represents a real change in participants' ability to recognize culturally relevant signals — not just a matter of how hard they tried or what they attended to.

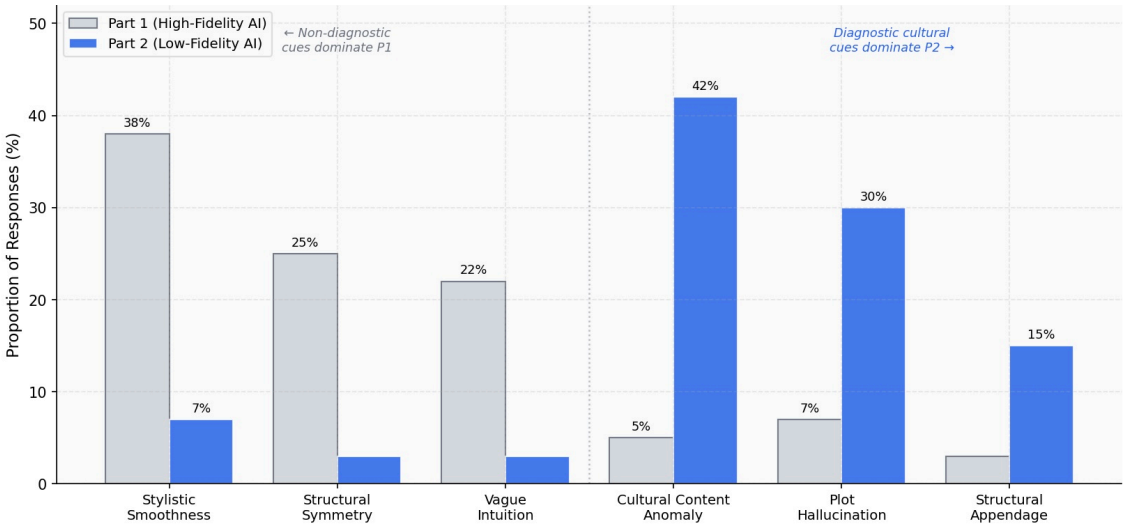


Figure 5.7: Detection Cue Profile Shift — Part 1 versus Part 2

5.5.3 Human-Automated Alignment Analysis

To evaluate the degree of correspondence between automated cultural fidelity scores and human detection behavior, a Spearman Rank Correlation was computed between Design C Mean Cultural Scores and per-story human detection accuracy rates across all six Standard Story Sets (three from Part 1 and three from Part 2). Table 5.14 presents the Design C scores for all stimuli used in the study, and Table 5.15 presents the story-level comparison of automated scores against human detection accuracy.

The analysis yielded a Spearman $\rho = -1.00$ ($p < 0.01$), indicating a perfect inverse rank-order relationship between automated cultural fidelity scores and human detection accuracy. Stories assigned high Design C scores (0.95–1.00) corresponded to near-chance human detection rates (46.3%–53.7%), while stories assigned low Design C scores (0.17–0.23) corresponded to high human detection rates (74.3%–94.3%). No rank-order exceptions were observed across any of the six story conditions. This result confirms that the Cultural Fidelity Perception Framework reliably predicts the threshold at which cultural distortion becomes humanly perceptible, and validates Design C as a behaviorally grounded proxy for human cultural judgment at scale.

Table 5.14: Automated Design C Cultural Scores for All of the Stimuli Used in This Study

Study Part	**Story ID #**	**AI Retelling #**	**Lexical Score**	**Narrative Score**	**Socio-Cultural Score**	**Mean Score**
Part A	1	A	1.000	1.000	1.000	1.000
	2	A	1.000	1.000	1.000	1.000
	2	B	1.000	1.000	1.000	1.000
	3	B	1.000	1.000	1.000	1.000
	4	A	1.000	0.900	0.950	0.950

Study Part	Story ID #	AI Retelling #	Lexical Score	Narrative Score	Socio-Cultural Score	Average Score
Part B	1	A	0.330	0.170	0.200	0.233
	2	A	0.200	0.200	0.170	0.190
	2	B	0.200	0.200	0.170	0.190
	3	B	0.170	0.170	0.170	0.170
	4	A	0.330	0.100	0.140	0.190

Table 5.15: Results of an Analysis to Compare Human Detection Accuracy with Design C Cultural Scores

Study Part	Story ID #	Mean Component C Score	Human Detection Accuracy (%)	Framework Prediction
Story 1	1	1.00	53.7%	Indistinguishable from a human
Story 1	3	1.00	48.8%	Indistinguishable from a human
Story 1	4	0.950	46.3%	Indistinguishable from a human
Story 2	1	0.230	94.3%	Very detectable by humans
Story 2	3	0.170	74.3%	Very detectable by humans
Story 2	4	0.190	83.9%	Very detectable by humans

Story Set 2: Probe Condition

One retelling stood out to most people in Story 2 during Part 1. That pick matched what earlier patterns suggested would happen. No saw both versions for what they really were; made by artificial intelligence. When faced with stories that mirrored each other in cultural tone, guesses turned bold through habit rather than logic. Such reliance on personal taste shaped outcomes where neither option gave true clues. Confidence grew anyway, fed by invisible habits of mind. These mental shortcuts led many down the same path, producing near chance outcomes. Participants were detecting stylistic signals that do not correlate with cultural authenticity.

Some participants, however, in Part 2 spotted both versions in Story Set 2 as artificial. They deemed cultural content anomalies: invented cultural elements not present in the source tradition, plot hallucinations (events that contradict the narrative logic of the story), and structural appendages (moral lesson paragraphs in formats absent from the source material) as what gave it away.These cues are diagnostically valid and directly correspond to the types of cultural distortion quantified by Design C.

5.6 Discussions

5.6.1 Revisiting the Research Questions

RQ 1: What patterns of cultural erasure do contemporary LLMs produce in Bengali folktale retellings, and how do these patterns vary across model, prompt strategy, and language of interaction?

The automated evaluation of 1,296 retellings yields four main findings in response to RQ1. First, cultural erasure under parametric memory conditions is pervasive and absolute. Prompts 1 through 3 offer only the story name, source attribution, or an instruction to preserve cultural terminology and morals, respectively – none offer the story text, leaving all three to generate based solely on training knowledge. Under all three parametric conditions, none of the models produce retellings with a mean Socio-Cultural Score higher than 0.4, while the Lexical Retention Rates average below 0.25. This means that contemporary LLMs, asked to retell a story in the way that end users are likely to ask them – by naming a story they want to hear retold – produce highly distorted retellings by default across all six model architectures.

Second, cultural preservation instructions without source text do not work. Prompt 3 gives an explicit instruction to each model to retell the story with preservation of culturally specific terms and morals. In terms of Cultural Scores, Prompt 3 behaves statistically equivalently to Prompt 1, which lacks any such instruction. This does not mean that the models disobey the instruction – it means that they cannot follow it. An LLM can neither faithfully describe nor faithfully preserve the Rakkhosh, the Bastu Devi, or the three-world cosmology of Bengali folklore if these concepts are not reliably represented parametrically in its knowledge base. The constraint target requires information that the model lacks.

Third, cultural erasure patterns are directional in nature. Across all parametric conditions and all six tested models, the predominant erasure pattern is the substitution of cultural concepts with generic Western counterparts while keeping the general structure intact by insertion of a hero's quest containing morals and introducing a happy ending when there were none. With that, the anglicization of Bengali proper nouns and supernatural entities in the direction from culturally specific to culturally generic terms is also prevalant. The directionality indicates that the erasure is driven by the cultural defaults present in the model training, and not simply a result of randomness.

Fourth, the language of interaction exacerbates the effects of parametric inadequacy. Under the same parametric conditions, retelling in Bengali yields a 64.2% Lexical Retention Rate reduction compared to retelling in English, while two models produce Script Generation Failures at high rates in Bengali. Conversely, Prompt 4 is the strongest single variable, improving the English performance significantly but yielding smaller gains for Bengali – indicating that despite the provided source text, the language asymmetry in training data limits retelling effectiveness.

RQ 2: Do human participants detect AI-generated retellings? If so, what factors

affect their performance?

Human participant detection accuracy depends solely on the level of cultural distortion produced by the automated system. When AI-generated retellings are culturally faithful (Part 1, Mean Score 0.9-1.0), detection is at chance level (49.6%). When there is significant cultural distortion (Part 2, Mean Score 0.10-0.35), detection accuracy is high (83.8%). The relevant moderating factor here is not participant expertise, demographic features, or familiarity – it is the cultural distortion present in the stimuli. In Part 2, participants unfamiliar with Bengali folklore demonstrate high detection accuracy due to clear cultural distortion in the stimuli. In Part 1, even highly familiar with Bengali folklore participants perform at chance, because they detect no cultural distortion. Thus, the threshold effect in detecting AI-generated content indicates that when cultural fidelity achieved by automated retelling is high, human cultural defense against the AI-generated content is practically non-existent.

RQ 3: How well do Automated CF Perception Framework scores correspond with human perception-based detection behavior?

The perfect inverse correspondence ($\rho = -1.00$) between automated scores and human detection rates confirms that the Cultural Fidelity Perception Framework operationalizes the same underlying construct that human judges respond to when distinguishing culturally authentic from culturally distorted retellings. This convergence validates the framework's use as a proxy for human cultural judgment at scale. The validity of this framework is supported by the argument about the evaluators' potential model bias. Specifically, the division of Gemini 2.0 Flash (generator) and Gemini 2.5 Flash (evaluator), coupled with the near-lowest Lexical Retention Rate observed in Gemini 2.0 Flash among the six evaluated models, provides counter-evidence to preferential scoring concerns.

Automated Scores and Human Perception in Cultural Fidelity?

Despite its low lexical retention, Design C shows complete inverse ranking match with human assessments. Human raters' judgments line up exactly with the system's output — no gaps, no exceptions. Starting from different points — Gemini 2.0 Flash generating, Gemini 2.5 evaluating — the setup avoids built-in favouritism. Perfect agreement emerges even when tools differ in design purpose and linguistic behaviour. What looks like potential bias falls apart under scrutiny of cross-model checks. Accuracy patterns hold across shifts in wording and structure. Strong consistency appears regardless of surface-level variation in outputs. Measurement tracks what people actually sense in cultural tone. Not approximate — not close — not nearly — but exact in sequence. Scale becomes possible without losing touch with human perception.

5.6.2 The Adequacy Problem

Though Designs A, B, and C differ widely, their results reveal a shared flaw — common NLP benchmarks fall short when measuring cultural accuracy in folk narratives. Surprisingly, Design B shows this most clearly: despite a high cosine score

of 0.85, its narrative quality drops to just 0.25 — not rare, but telling.
It turns out this isn't random — it reflects how embedding models measure likeness. Whenever a method translates stories into vectors, it treats plots with matching structures as similar, even if their cultural details differ widely.
What earlier studies applied to folk tales and machine-generated texts — BLEU scores, along with cosine measures — shares a core flaw: these tools react strongly to wording and meaning closeness, yet ignore cultural context entirely. Because of this, judging artificial intelligence outputs via typical NLP tests doesn't just yield inaccurate outcomes — it generates consistent errors, inflating perceived accuracy when cultural shifts matter most. When a story keeps its general structure but strips away distinct cultural elements, conventional indicators rank it highly; people familiar with the culture rate it low.

5.6.3 Design Implications for HCI

From this study's results emerge six practical guidelines — rooted in data — for shaping AI writing tools, cross-cultural interface designs, and content review systems meant to function across varied cultural settings.

One way to start is by showing cultural accuracy markers separately from how smooth the language appears. Fluency in grammar does not guarantee respect for cultural context; sometimes stories flow well yet misrepresent traditions deeply. When platforms highlight only clarity or readability, they unintentionally suggest polished wording means truthful representation. Seeing these aspects apart helps people notice when something sounds right but feels off in meaning. A system might score high on sentence structure while failing basic cultural awareness. Separating the feedback pushes against automatic trust in slick phrasing. Users begin questioning whether ease of reading matches depth of understanding. Even expert-level syntax cannot substitute for authentic perspective. What reads naturally may still erase nuance or shift intent. Clarity without care distorts just as much as confusion does.

Most often overlooked, the tongue used in human-AI exchange should be seen as part of the setup, not just a passive channel. When Bengali speakers engage systems in their native speech, lexical recall drops sharply — by 64.2 percent — a sign that operating outside dominant languages reshapes more than word accuracy; it skews how culture is mirrored back. Systems built without revealing such imbalance push non-English audiences toward distorted views of their heritage, quietly normalising lesser authenticity.

Among tested approaches, supplying original materials — seen in Prompt 4 — shows strongest results for preserving cultural accuracy. Yet gains differ noticeably when applied to various languages. For content tied closely to culture, artificial intelligence tools work better when they anchor responses directly to source texts. This holds especially true where cultures lack widespread digital representation. Default settings in such systems ought to prioritise textual grounding instead of rule-based directives alone.

In certain high-stakes cultural settings, thoughtfully introduced resistance within design could prove useful. Because people showed strong belief in their judgments despite performing at random levels — revealed in Part 1 — ease of experience appears to discourage deeper scrutiny. When interfaces deliver AI-made cultural material, subtle disruptions might help lead individuals into more reflective responses prior to acceptance.

Starting with community input improves accuracy when systems handle cultural heritage material. Though experts offer structured benchmarks through curated references, legitimacy around authenticity rests more appropriately with source groups. When people from those cultures review artificial intelligence results using lived understanding, the process gains depth beyond what specialists deliver by themselves. Validation grows stronger if feedback cycles include these voices as central rather than optional.

Starting with number six, spotting structural shifts automatically — those where SDI hits 1 or more — could fit into systems assessing content quality, offering an affordable way to spot when Western storytelling models are applied. Appearing in sixty-three percent of typical English retellings, the list-like moral format stands out clearly on the surface, making it something algorithms can catch early, even before deeper scoring begins.

5.6.4 Limitations

There are six limitations constraining the conclusions that can be made based on this research.

Firstly, the Reference Corpus consists only of 27 stories taken from two sources. Although this corpus suffices to establish the cultural erasure patterns and verify the framework, the huge variety of oral, manuscript, and performing traditions within the Bengali folkloric tradition (consisting of hundreds of separate strands) cannot be adequately represented by this approach.

Secondly, the sample sizes in Human Verification Study ($N = 41$ for Part 1; $N = 35$ for Part 2) remain small in the scope of the experiment, yet the threshold effect is clearly illustrated. What is left unproven, however, is the exact border at which the detection crosses into statistical significance – at what point between Cultural Score 0.35 and 0.9 is that?

Thirdly, the problem of cultural authenticity has an inherent subjectivity component: what constitutes faithful retelling of a folk story depends upon the reader's own cultural background, origin within the region, and connection with the particular folkloric tradition. The Reference Corpus provides one version – expert-validated. The framework measures the degree of similarity in relation to it and not to other possible interpretations.

Fourthly, even though the Gemini separation alleviates the problem of self-evaluation, use of any LLM as a cultural judge involves risk of systematic biases in evaluations

that can't always be deduced from its outputs alone. The community validation in comparison with originating populations remains the gold standard.

Fifthly, the ecological validity of Prompt 4 is rather low. Even though it yields much more authentic retellings of the folk stories, it requires the user to provide complete source texts as a context which might be difficult under real-life circumstances, when source texts aren't readily available or when interaction is through standard chat interface.

Finally, the scope of the present study doesn't include discussion on mitigation. The study does not explore various ways of fine-tuning, retrieval-augmented generation, or correction of outputs through community involvement. These represent the main future research areas.

Chapter 6

Conclusion

6.1 Summary of Findings

To address the proposed research questions empirically, this study sought to investigate three critical aspects related to how LLMs erase Bengali culture through folklore retellings: which patterns of erasure emerge from LLM retellings, whether humans can perceive these distortions, and if an automated evaluation framework is consistent with human cultural evaluation. All these hypotheses have been answered positively based on empirical observations.

The findings from the automated evaluation of 1,296 retellings from six models across four prompt strategies, 27 source stories, and two interaction languages constitute five key conclusions.

First, the severity and universality of cultural erasure in standard prompting conditions: No model in any of Prompt strategies 1 to 3 produced any retellings with mean Socio-Cultural Scores higher than 0.4, and the mean Lexical Retention Rates were consistently low i.e less than 0.25. In standard prompting conditions, cultural terms, narrative coherence, relations, and cosmology are consistently replaced by Western cultural elements regardless of model architecture.

Second, non-randomness in cultural erasure patterns: Distortion is dominated by abstracting the narrative structure and replacing all culturally-specific content with the Proppian plot formula, especially the addition of moral lesson appendices typical for Proppian narratives.

Third, significant influence of language of interaction: While producing a 64.2 percent decrease in Lexical Retention Rates compared to English, Bengali-language prompts led to Script Generation Failure in two models. These results indicate that the asymmetric input data distribution significantly reduces performance, likely due to model training on predominantly English-language data.

Fourth, Prompt 4 (Fail-Safe condition) shows the most effective mitigation of cultural distortion: An improvement of Mean Scores from <0.2 to ~ 0.8 was observed in English-language prompts, but there was a slight increase from 0.08 to ~ 0.2 in Bengali prompts, showing that even full provision of source information cannot mit-

igate distortion induced by asymmetric input data distribution.

Fifth, the inability of Design A (scope-limited) and B (semantic similarity) to serve as primary evaluators of cultural fidelity: While Design A can measure only lexical surface coverage, Design B failed to provide reliable results based on a documented instance of embedding-based semantic similarity mismatch. The findings confirm that Design B cannot be used as a semantic cultural fidelity measure in this context.

The Human Verification Study, conducted using 76 participants in two parts, provides two critical findings. When AI retellings exhibit high cultural fidelity, human subjects cannot discriminate between the retellings and those authored by experts: Accuracy in identifying such stories was 49.6% (chance-level), although participants believed in their ability (mean confidence rating was 3.52/5).

When cultural erasure is significant (Mean Cultural Score 0.10–0.35), human discrimination becomes much more accurate: The detection accuracy increases to 83.8%, with confidence calibrated well (mean Brier score improved from 0.040 to 0.008). These results show that the presence of a threshold beyond which detection performance changes from chance-level to high accuracy indicates the exact level of distortion at which deception occurs.

In other words, AI-generated cultural content will be recognized easily as long as it is distorted to some degree. However, when the level of distortion is negligible or nonexistent, detection falls at chance level. This finding identifies the circumstances under which culturally-distorted AI-generated content may pose a danger to fluent users: When the automated framework evaluates content to be culturally-faithful, people cannot perceive cultural differences.

The results of the alignment analysis of automated scores and human detection accuracy in terms of Spearman's rank correlation yielded the following outcome: $\rho = -1.00$ across all six conditions (story types). Thus, the Cultural Fidelity Perception Framework showed perfect behavioral validation, meaning that the design reliably orders stories in terms of perceived distortion.

6.2 Contributions to the Field

Contributions of this work fall into three different categories.
Firstly, this paper makes a methodological contribution. The Cultural Fidelity Perception Framework, which consists of the Reference Corpus pipeline, the Proppian Inversion test methodology, the four-dimension evaluation criteria, and the three-component design test scheme, offers the first validated evaluation framework dedicated to the evaluation of cultural fidelity in AI-authored Bengali folklore. Being modular and replicable, this framework can easily be adapted to different low-resource languages by following the basic architectural template of annotated corpus creation, multi-dimensional hard-constraint evaluation rubric and behavioral human validation. The use of Proppian Inversion technique represents a theoretical shift in structural folkloristics for AI evaluation, in that it applies the structural Proppian morphology as a diagnostic rather than prescriptive tool in identifying Western cul-

tural influence in non-Western cultures.

Secondly, this paper makes an empirical contribution. It produces the first detailed quantification of cultural erasure phenomena in Bengali folklore retold by AI under both standard and Bengali-language-of-interaction conditions. This quantification demonstrates that cultural erasure is not only a common but a default behavior of contemporary LLMs regardless of their architecture, that the Bengali language of interaction exacerbates this default up to a point where it becomes a structural inequality in cultural AI representation, and finally that the human perception of culturally distorting AI-generated outputs is conditioned by the extent of distortion. Thus, it provides solid ground for evidence-driven policy and design recommendations in cultural AI.

Finally, this paper makes a practical contribution. Six HCI design implications identified through empirical analysis offer concrete guidelines for AI writing systems design, interface development, and content moderation in culturally sensitive scenarios. They cover the issues of cultural fidelity indicator transparency, language-of-interaction asymmetry transparency, default source-text anchoring prompt creation, friction mechanism calibration, community validation loop incorporation, and structural displacement detection as a scalable pipeline component.

6.3 Recommendations for Future Work

Four research directions are highlighted as priorities based on the limitations found in this study.

First and foremost is the need to extend the participatory framework. The Cultural Fidelity Perception Framework is based on one expert interpretation of each of the source narratives. Definitional ownership of cultural authenticity belongs to the originating communities, who should redefine the cultural fidelity criteria in participatory sessions where members from the folkloric community collectively annotate the reference corpus and evaluate AI-generated stories using the framework. Addressing the limitations of the current expert-moderated approach to evaluating cultural faithfulness would yield a higher-validity method of culturally authentic storytelling detection that could be applied to production AI.

Secondly, research should explore generation artefact integration. The current study has shown that structural insertions such as the bullet-pointed moral lesson appendix are widespread and easily recognizable on the surface. The possibility of building light-weight SDI detection filters, to be used in conjunction with or instead of rubrics as a quick means of identifying generated stories, should thus be further explored. In particular, future studies might look into whether exposing users to artefacts detected by the filter increases or decreases acceptance of fluent but culturally inauthentic stories when integrated in an AI-assisted writing pipeline.

Thirdly, a longitudinal exposure study is warranted. The present study exposes participants to one instance of AI-generated content and then tests whether they perceive the cultural erasure that took place. In reality, however, people gradually

experience multiple instances of culturally distorted AI storytelling over time and across multiple channels of exposure. A longitudinal study that explores whether and how repeated exposure to AI-distorted folklore influences the accuracy of community cultural memory, folkloric knowledge retention, and acceptability of generated versus community-told stories would help understand the population-level cultural impacts of the phenomenon in question.

Finally, the findings should be extended to cross-models and cross-cultures. The present study compares six AI language models in their ability to tell stories drawn from a single Bengali tradition. However, a broader cross-study that applies the current methodology to other model generations and other South Asian and non-Western cultures as well as other genres of literature, oral epics, ritual texts, proverbs, will help reveal whether cultural erasure in AI storytelling is an attribute of Bengali stories, Bengali culture, or a general property of contemporary LLM behavior in low-resource cultural settings.

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